Dynamics and relativity

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Lecture 1 – 22/01/26

§1 The structure of the Newtonian universe

- Three dimensional space endowed with a *Cartesian reference frame* – an origin and some axes.
- Time can be labelled in an arbitrary reference frame, by a real number t.
- A *point particle* is an idealised object that is completely determined by its position at a given point in time, $\mathbf{x}(t)$. Its velocity

$$\mathbf{v} = \frac{d\mathbf{x}}{dt} = \left(\frac{dx_1}{dt}, \frac{dx_2}{dt}, \frac{dx_3}{dt}\right) = \dot{\mathbf{x}}$$

is tangent to the trajectory. The acceleration

$$\mathbf{a} = \frac{d^2\mathbf{x}}{dt^2} = \left(\frac{d^2x_1}{dt^2}, \frac{d^2x_2}{dt^2}, \frac{d^2x_3}{dt^2}\right) = \ddot{\mathbf{x}} = \dot{\mathbf{v}}$$

The above is *not* sufficient to write down Newton's equations. Consider, for example, a free particle not experiencing any forces. The position of this particle is $\mathbf{x}(t)$, but which frame should we use?

§1.1 Law of inertia

Proposition 1.1 (Law of inertia)
There exist *inertial frames* in which a free particle has a constant velocity.

This is an improved version of Newton's first law. Indeed, it is a true statement, but not an obvious one.

§1.2 Galilean relativity principle

Definition 1.2. A *Galilean transformation* has the form

$$\mathbf{x}' = R\mathbf{x} + \mathbf{k} + \mathbf{w}t$$

where $R \in O(3)$, $\mathbf{k}$ is a translation vector, and $\mathbf{w}$ is a *boost*. In particular, $\ddot{\mathbf{x}}' = \ddot{\mathbf{x}}$.

Proposition 1.3 (Galilean relativity principle)
A frame related to an inertial frame by a Galilean transformation is also an inertial frame, and all laws of physics are the same in both frames.

Example of a boost: dropping a ball on a boat moving at constant velocity.

Galilean invariance also restricts the forces that are possible (see example sheet). For example, there are no 'special points' in the universe (or special directions, times, velocities, etc) – they are all relative. This is *not* the case for acceleration – an object accelerating in one inertial frame is accelerating with the same magnitude in all inertial frames.

Galilean transformations form the *Galilean group*, often supplemented by shifts in time, under which the laws of physics are also invariant. In particular, this implies that all inertial frames have the same time, called absolute time.

§2 Forces

Once there is more than one particle in the universe, there will be interactions between them. In Newtonian physics, these are described by forces.

§2.1 Newton's second law

Theorem 2.1 (Newton's second law)
In an inertial frame,

$$\dot{\mathbf{p}} = \mathbf{F}$$

where the *momentum* $\mathbf{p}$ is $m\mathbf{v}$, where m is the *inertial mass*, mass for short.

The mass is an additional property for point particles that we take to be constant in time unless otherwise stated.

The force $\mathbf{F}$ depends on the ongoing interactions, but it can only depend (axiomatically) on $\mathbf{x}$ and $\mathbf{v}$. This implies that Newton's second law is a second-order differential equation, and that, given $\mathbf{x}$ and $\mathbf{v}$ at $t = 0$ for all particles, Newton's equations will uniquely determine $\mathbf{x}(t)$ for all future times.

We should note that Newtonian mechanics has been superseded by quantum mechanics (for small things) and general relativity (for fast things).

§2.2 Conservative forces

Definition 2.2. A force $\mathbf{F}$ is called *conservative* if it can be written

$$\mathbf{F} = -\boldsymbol{\nabla} V$$

for some *potential* scalar field V.

For example, *gravitational potential energy* of a particle of mass m at $\mathbf{x}$ due to a particle of mass M at $\mathbf{x}_0$ is

$$V = -\frac{GMm}{|\mathbf{x}-\mathbf{x}_0|}$$

where G is a constant.

Now we can get the force by taking the gradient. First,

$$\partial_i\left(|\mathbf{x}-\mathbf{x}_0|^2\right) = 2|\mathbf{x}-\mathbf{x}_0|\partial_i|\mathbf{x}-\mathbf{x}_0|$$

But also,

$$\begin{aligned}\partial_i\left(|\mathbf{x}-\mathbf{x}_0|^2\right) &= \partial_i\left((\mathbf{x}-\mathbf{x}_0)_j(\mathbf{x}-\mathbf{x}_0)_j\right)\\ &= 2(\mathbf{x}-\mathbf{x}_0)_j\partial_i(x_j-(x_0)_j)\\ &= 2(\mathbf{x}-\mathbf{x}_0)_j\delta_{ij}\\ &= 2(\mathbf{x}-\mathbf{x}_0)_i\end{aligned}$$

and so we have

$$\begin{aligned}\partial_i|\mathbf{x}-\mathbf{x}_0| &= \frac{(\mathbf{x}-\mathbf{x}_0)_i}{|\mathbf{x}-\mathbf{x}_0|}\\ \implies \boxed{\boldsymbol{\nabla}|\mathbf{x}-\mathbf{x}_0| = \frac{\mathbf{x}-\mathbf{x}_0}{|\mathbf{x}-\mathbf{x}_0|}}\end{aligned}$$

Hence

$$\mathbf{F} = -\boldsymbol{\nabla} V = -GMm\frac{\mathbf{x}-\mathbf{x}_0}{|\mathbf{x}-\mathbf{x}_0|^3}$$

Let $\mathbf{r} = \mathbf{x}-\mathbf{x}_0$, then

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is a unit vector. Thus we have the inverse square law of gravity.

> **Remark.** Sometimes we write $V = m\varphi$ where $\varphi = -\frac{GM}{|\mathbf{x}-\mathbf{x}_0|}$ is called the *gravitational potential.*

Near the surface of the Earth, take $\mathbf{x}_0 = \mathbf{0}$, and $|\mathbf{x}| = R+z$, where R is the radius of the Earth and $z \ll R$. Then

$$\varphi(R+z) = -\frac{GM}{R+z}$$

$$
\begin{aligned}
&= -\frac{GM}{R}\left[1 - \frac{z}{R} + \frac{z^2}{R^2} + \cdots\right] \\
&\approx \text{constant} + \underbrace{\frac{GM}{R^2}}_{g} z
\end{aligned}
$$

where $g \approx 9.8\text{ms}^{-2}$. Since the constant will drop out when taking the gradient, we ignore it, so we have

$$
\begin{aligned}
&\varphi \approx gz \\
\implies &\mathbf{F} = -mg\hat{\mathbf{z}}
\end{aligned}
$$

This force leads to the simplest non-trivial example of motion due to a force. By Newton's second law, we have

$$m\ddot{\mathbf{x}} = m\mathbf{g}$$

where $\mathbf{g} = (0, 0, -g)$. We need only consider the z component, which gives

$$
\begin{aligned}
&m\ddot{z} = -mg \\
\implies &\dot{z} = v_0 - gt \\
\implies &z = z_0 + v_0 t - \frac{1}{2}gt^2
\end{aligned}
$$

for an initial height z_0 and an initial velocity v_0.

§2.3 Energy

Proposition 2.3

Conservative forces have a *conserved energy*

$$
\begin{aligned}
E &= \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + V(\mathbf{x}) \\
&= \frac{1}{2}m\dot{x}_i\dot{x}_i + V(\mathbf{x})
\end{aligned}
$$

Proof. We have

$$
\begin{aligned}
\frac{dE}{dt} &= m\dot{x}_i\ddot{x}_i + \frac{\partial V}{\partial x_i}\dot{x}_i \\
&= \dot{x}_i\left(m\ddot{x}_i + \frac{\partial V}{\partial x_i}\right) \\
&= 0 && (m\ddot{\mathbf{x}} = -\boldsymbol{\nabla}V)
\end{aligned}
$$

■

Example 2.4

Suppose we throw an object into space and want it to never fall back down. The minimal velocity required is called the escape velocity.

When the object is thrown, its energy is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

To not fall back, the object must reach $|\mathbf{x}| \to \infty$ without the velocity going to zero. We have

$$E_\infty = \frac{1}{2}mv_\infty^2 - 0$$

By conservation of energy, $E = E_\infty$, and so to make $v_\infty > 0$ we must have

$$\begin{aligned} \frac{1}{2}mv^2 &> \frac{GMm}{R} \\ \implies v^2 &> \frac{2GM}{R} \\ \implies v &> \sqrt{\frac{2GM}{R}} \end{aligned}$$

which is approximately 10kms^{-1} for Earth.

Note: the mass cancels in v_{escape} because the *gravitational mass* (from the inverse square law) is equal to the *inertial mass* (that appears in Newton's second).

Proposition 2.5 (Principle of equivalence)

Objects have the same gravitational and inertial masses.

Remark. This is not known and there is no reason why we would expect this to be true – whether there may be tiny deviations between the masses is under investigation.

It's useful to write energy as $E = T + V$, where T is the *kinetic energy* $\frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}$.

Definition 2.6. The *work done* by a force as a particle moves along a trajectory C is given by

$$W = \int_C \mathbf{F} \cdot d\mathbf{x}$$

Proposition 2.7

For a conservative force, the work done along a trajectory only depends on the endpoints of the trajectory, not on the path itself.

Proof 1. We have

$$W = \int_C \mathbf{F} \cdot d\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F} \cdot \frac{d\mathbf{x}}{dt}\, dt$$

$$\begin{aligned}
&= m \int_{t_1}^{t_2} \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}}\, dt \\
&= \frac{1}{2} m \int_{t_1}^{t_2} \frac{d}{dt} (\dot{\mathbf{x}} \cdot \dot{\mathbf{x}})\, dt \\
&= T(t_2) - T(t_1) \\
&= V(t_1) - V(t_2) && (T_2 + V_2 = T_1 + V_1) \\
&= V(\mathbf{x}(t_1)) - V(\mathbf{x}(t_2)) \\
&= V(\mathbf{x}_1) - V(\mathbf{x}_2)
\end{aligned}$$

from which it's obvious that the work done only depends on the endpoints. ■

Proof 2. We have

$$\begin{aligned}
W = \int_C \mathbf{F} \cdot d\mathbf{x} &= -\int_C \boldsymbol{\nabla} V \cdot d\mathbf{x} \\
&= V(x_1) - V(x_2)
\end{aligned}$$

■

§2.4 Electromagnetic forces

Forces that depend on velocity typically don't have a conserved energy (e.g. friction), but the Lorentz force is an exception.

Electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ exert the following force on a particle with *charge* q:

$$\mathbf{F} = q \left[\mathbf{E}(\mathbf{x}) + \dot{\mathbf{x}} \times \mathbf{B}(\mathbf{x})\right]$$

We will restrict to static electromagnetic fields, in which case we have

$$\mathbf{E} = -\boldsymbol{\nabla} \varphi$$

for some electric potential φ. We claim the conserved energy is $E = \frac{1}{2} m \dot{\mathbf{x}}^2 + q\varphi(\mathbf{x})$. We have

$$\begin{aligned}
\frac{dE}{dt} &= m\ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} + q\boldsymbol{\nabla}\varphi \cdot \dot{\mathbf{x}} \\
&= (\mathbf{F} + q\boldsymbol{\nabla}\varphi) \cdot \dot{\mathbf{x}} \\
&= q(\dot{\mathbf{x}} \times \mathbf{B}) \cdot \dot{\mathbf{x}} \\
&= 0
\end{aligned}$$

The velocity-dependent force is orthogonal to the trajectory of the particle, so it does no work on the particle and energy is still conserved.

Electric forces are similar to gravitational ones: the potential φ at $\mathbf{x}$ due to another particle of charge Q at $\mathbf{x}_0$ is

$$\varphi = \frac{Q}{4\pi\varepsilon_0} \cdot \frac{1}{|\mathbf{x} - \mathbf{x}_0|}$$

where ε_0 is the *permittivity of free space*. Just like with gravity, this leads to an inverse square law called the *Coulomb force*.

An important distinction from gravity is that charges can be positive or negative, but mass is always positive.

Magnetic forces are different. A charged particle in a magnetic field obeys

$$m\ddot{\mathbf{x}} = q\dot{\mathbf{x}} \times \mathbf{B}$$

This is a vector differential equation, which we can solve by writing out in Cartesian coordinates (we will also take $\mathbf{B}$ constant and in the z-direction, $\mathbf{B} = (0, 0, B) = B\hat{\mathbf{z}}$). The equations become

$$\begin{aligned} m\ddot{x} &= q\dot{y}B && (1) \\ m\ddot{y} &= -q\dot{x}B && (2) \\ m\ddot{z} &= 0 && (3) \end{aligned}$$

The first thing we can observe is that $z = z_0 + v_z t$, that is, the particle moves with constant velocity in the z-direction. Now, differentiating (1) and subbing into (2), we have

$$m\dddot{x} = q\ddot{y}B = -\frac{q^2\dot{x}B^2}{m}$$

and this is a second-order equation for $\dot{x}$ – we have

$$\dot{x} = \tilde{A}\sin(\omega t + \varphi)$$

where $\omega = \frac{qB}{m}$ is the *cyclotron frequency*. Hence

$$\boxed{x = x_0 + A\cos(\omega t + \varphi)}$$

Plugging into (1), we have

$$\begin{aligned} qB\dot{y} &= -mA\omega^2\cos(\omega t + \varphi) \\ \implies &\boxed{y = y_0 - A\sin(\omega t + \varphi)} \end{aligned}$$

Hence, the particle is moving in a circle in the x-y plane while moving with a constant velocity in the z-direction.

DIAGRAM.

Alternatively, let $\xi = x + iy$, and write $(1) + i(2)$. We have

$$m\ddot{\xi} = -iqB\dot{\xi}$$

which we can solve nice and easily: $\xi = c_1 e^{-i\omega t} + c_2$. Let's put

$$\begin{aligned} c_1 &= Ae^{-i\varphi} \\ c_2 &= x_0 + iy_0 \end{aligned}$$

Plugging in and taking real and imaginary parts, we recover the exact same solution as before.

> **Remark.** There is something 'complex' underlying cyclotron motion – c.f. quantum Hall effect.

§2.5 Motion in one dimension

Often, problems involving higher dimensions can be reduced to one-dimensional motion. In such a case, we have

$$m\ddot{x} = F_x$$

If F_x is independent of velocity, it can always be written as the gradient of a potential by setting

$$V(x) = -\int_{x_0}^{x} F_x(x')dx'$$

for an arbitrary reference point x_0. So the energy

$$E = \frac{1}{2}m\dot{x}^2 + V(x)$$

is always conserved. Then, putting $E =$ constant, we get a first-order differential equation for $x(t)$ which is easily integrated: we have

$$\begin{aligned} \dot{x} &= \pm\sqrt{\frac{2}{m}(E - V(x))} \\ \implies t - t_0 &= \pm\int_{x_0}^{x} \frac{1}{\sqrt{\frac{2}{m}(E - V(x'))}}dx' \end{aligned}$$

and so solving Newton's equations in one dimension essentially just amounts to doing this integral.

Lecture 4 – 29/01/26

From

$$E = \frac{1}{2}\dot{x}\cdot\dot{x} + V(x)$$

we conclude that $E > V(x)$. This restricts the range of where the particle can possibly be. The points where $E = V(x)$ and $\dot{x} = \mathbf{0}$ are called *turning points*, since the particle typically 'bounces off' the potential barrier at these points. Depending on whether there are potential barriers on all sides or just a few, the motion of the particle may be *bounded* or *unbounded*.

A special case occurs when $E = V(x)$ and $V'(x) = 0$. These are called *equilibrium points* – the particle can sit indefinitely. At such a point, we have

$$\begin{aligned} m\ddot{x} &= -V'(x) = 0 \\ E &= \frac{1}{2}m\dot{x}^2 + V(x) \\ \implies &\boxed{\dot{x} = \ddot{x} = 0} \end{aligned}$$

Furthermore, motion near an equilibrium point is especially simple. Let x_0 be an equilibrium point, and Taylor expand:

$$V(x) \approx V(x_0) + \underbrace{(x - x_0)V'(x_0)}_{0} + \frac{1}{2}(x - x_0)^2 V''(x_0)$$

If $V''(x_0) > 0$, this is the potential for a harmonic oscillator. We end up with

$$\begin{aligned} m\ddot{x} &= -V'(x) = -(x - x_0)V''(x_0) \\ \implies x &= x_0 + A\cos(\omega t + \varphi) \end{aligned}$$

so we have oscillations with frequency $\omega = (V''(x_0)/m)^{1/2}$.

If the amplitude A is small enough, then we can neglect higher-order terms in the Taylor expansion and the solution is self-consistent (we won't end up getting too far from the potential well). This is called a *stable equilibrium point.*

If instead $V''(x_0) < 0$, we have an unstable equilibrium point:

$$x = x_0 + Ae^{\gamma t} + Be^{-\gamma t}$$

where $\gamma = \sqrt{-V''(x_0)/m}$. If $A \neq 0$, then we will move far from the equilibrium point and so the Taylor expansion will break down.

The case where $A = 0$ corresponds to the particle having just enough energy to reach a maximum of the potential somewhere else.

If $V''(x_0) = 0$, we simply go to a higher order of the Taylor expansion.

§2.6 Dimensional analysis

Dimensional analysis is a way to get information about solutions to equations without actually solving them.

At a mathematical level, it is the ability to rescale variables to remove certain constants from equations.

Consider the equation for a pendulum,

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin\theta$$

and let us attempt to find the period of θ. First, we substitute $t = \sqrt{\frac{l}{g}}\tau$. We then have

$$\begin{aligned} \theta(t(\tau)) &= F(\tau) \\ \frac{d^2F}{d\tau^2} &= -\sin F \end{aligned}$$

Clearly now the solution must be some function $f(\tau)$ – it cannot possibly depend on g and l. We conclude that the period $\Delta\tau$ satisfies

$$\begin{aligned} F(\tau) &= F(\tau + \Delta\tau) \\ \implies F\left(\sqrt{\frac{g}{l}}t\right) &= F\left(\sqrt{\frac{g}{l}}t + \Delta\tau\right) \\ &= F\left(\sqrt{\frac{g}{l}}\left(t + \sqrt{\frac{l}{g}}\Delta\tau\right)\right) \\ &= \theta\left(t + \sqrt{\frac{l}{g}}\Delta\tau\right) \end{aligned}$$

and so, without ever solving the eqation, we conclude that the period of a pendulum is proportional to the square root of its length.

In general the necessary rescaling may not be obvious. By endowing our constants and variables with so-called 'dimensions', we can perform this sort of analysis much more easily. Our basic dimensions are

- *length*, L;
- *time*, T;
- and *mass*, M.

Then we have results such as

$$\begin{aligned}[\dot{x}] &= \mathrm{LT}^{-1} \\ [\ddot{x}] &= \mathrm{LT}^{-2} \\ [F] &= \mathrm{MLT}^{-2} \\ [E] &= \mathrm{ML}^2\mathrm{T}^{-2}\end{aligned}$$

There may be other dimensions to keep track of, depending on the problem.

Theorem 2.8

For all physical laws, the dimensions of the left and right hand sides are equal.

Corollary 2.9

All arguments of non-trivial functions, i.e. those involving sums of different powers, must be dimensionless.

Example (Pendulum revisited)

We have $[g] = \mathrm{LT}^{-2}$, $[l] = \mathrm{L}$, $[m] = \mathrm{M}$, $[\theta_0] = 1$. We want the period T to have $[T] = \mathrm{T}$. So we let

$$\begin{aligned}T &= f(\theta_0)g^A l^B m^C \\ \implies [T] &= [g]^A[l]^B[m]^C \\ &= \mathrm{L}^A\mathrm{T}^{-2A}\mathrm{L}^B\mathrm{M}^C\end{aligned}$$

and we end up with the system

$$\begin{aligned}0 &= C \\ 0 &= A + B \\ 1 &= 2A\end{aligned}$$

which gives $T = f(\theta_0)\sqrt{\frac{l}{g}}$, as before.

If A, B, C are not all fixed, we have a dimensionless ratio and it will be necessary to apply some more thought.

Lecture 5 – 31/01/26

§2.7 Friction

Friction is not a fundamental force of nature, but a force that arises from complicated interactions that we don't want to consider in full detail.

In particular, an object passing through a medium such as air or water experiences microscopic forces between it and the medium – these forces cause momentum to be lost into the medium.

In part because of this, friction

- does not conserve energy (of the object);
- is *irreversible* – energy is lost by the object and never gained. This is closely related to the second law. In particular, the sign of a friction force must be such that it slows the object down. This implies, among other things, that friction depends on velocity.

Let us consider two important cases:

① *Linear drag*. We have $\mathbf{F} = -k_1\mathbf{v}$.

② *Quadratic drag*. We have $\mathbf{F} = -k_2|\mathbf{v}|\mathbf{v}$.

where k_1 and k_2 are called *coefficients of friction*.

Quadratic drag is more intuitive since collisions per unit time and momentum lost in each collision are both proportional to the particle's velocity.

Furthermore, the number of collisions should be proportional to the density of the medium, as well as the cross-sectional area of the object, so the force should also be proportional to these. Now, we can provisionally check units:

$$[F] = \mathrm{MLT}^{-2}$$
$$[k_2 v^2] = \mathrm{ML}^{-3}\mathrm{L}^2\mathrm{L}^2\mathrm{T}^{-2}$$

and these are equal, suggesting we have probably accounted for the main factors influencing k_2.

What about linear drag? Linear drag depends on viscous effects – the fact than an object drags the medium along with it as it moves and, as we move closer to the object, the velocity of the medium approaches v. It can then be shown that the momentum loss here is in fact linear in the velocity.

> **Example** (Stokes' law)
> For a spherical object, we have linear drag, and the coefficient is
>
> $$k_1 = 6\pi\eta L$$
>
> where η is the viscosity.

In general, we see both linear and quadratic drag, and which one dominates depends on the ratios of certain quantities. For example, taking k_1 equal to the value for Stokes' law above, we have

$$\frac{F_{\text{quad}}}{F_{\text{lin}}} \sim \frac{\rho A v^2}{\eta L v} \sim \frac{\rho L v}{\eta} = R$$

and this R is called the *Reynolds number*.

We are also yet to come onto *dry friction*, in other words the friction between two rigid bodies.

§2.7.1 Terminal velocity

Consider a particle falling under gravity with quadratic drag. For the z-component, we have

$$m\frac{dv}{dt} = -mg + kv^2$$

When the velocity is constant, the left hand side is zero, so we have

$$\begin{aligned} mg &= kv^2 \\ \implies v_{\text{term}} &= \sqrt{\frac{mg}{k}} \end{aligned}$$

Unlike escape velocity, we still have a mass here, so heavier objects reach a larger terminal velocity.

We can solve this equation analytically, but we can ask simple questions like: on what timescale does the object reach its terminal velocity? Since this should only depend on mass, gravity and k, we can apply dimensional analysis:

$$\begin{aligned} t &\propto m^A g^B k^C \\ \implies \mathrm{T} &= \mathrm{M}^A \mathrm{L}^B \mathrm{T}^{-2B} \mathrm{M}^C \mathrm{L}^{-C} \\ \implies &\begin{cases} A = -\frac{1}{2} \\ B = -\frac{1}{2} \\ C = \frac{1}{2} \end{cases} \end{aligned}$$

and so

$$t_0 = \sqrt{\frac{m}{kg}}$$

Hence, we know that v as a function of time takes the form

$$v = v_{\text{term}} F\left(\frac{t}{t_0}\right)$$

and so t_0 is the timescale on which the object reaches its terminal velocity.

We should also note that we can have terminal velocities arise from movement in the full 3 dimensions, giving the vector differential equation

$$m\frac{d\mathbf{v}}{dt} = m\mathbf{g} - k|\mathbf{v}|\mathbf{v}$$

§2.7.2 Damping

Friction also arises as a damping force in motion around an equilibrium point. For small oscillations, the linear drag is typically more important, giving the equation (in one dimension)

$$\ddot{x} = -\omega_0^2 x - 2\alpha\dot{x}$$

We have the solution

$$x = e^{-\alpha t}\left(A_+ e^{i\Omega t} + A_- e^{-i\Omega t}\right)$$

where $\Omega = \operatorname{Re}\left(\sqrt{\omega_0^2 - \alpha^2}\right)$. We have three cases:

- $\omega_0^2 > \alpha^2$: *Underdamping*. Here we have decaying oscillations
- $\omega_0^2 < \alpha^2$: *Underdamping*. Here we have exponential decay.
- $\omega_0^2 = \alpha^2$: *Critical damping*. Here we have a repeated root and instead get $x = (A + Bt)e^{-\alpha t}$.

> **Remark.** x and $\ddot{x}$ are invariant under *time reversal*, $t \to -t$. But $\dot{x}$ is not. As far as we know, all laws of nature are invariant under *CPT*, or charge-parity-time, symmetry, i.e. taking $q \to -q$, $\mathbf{x} \to -\mathbf{x}$ and $t \to -t$. But friction forces are odd under time reversal (and lack charge to compensate), so they are not fundamental.

§3 Central forces

An important class of potentials (gravity, electrostatics) only depend on the distance to the origin (or some other point) – we have

$$V(\mathbf{x}) = V(|\mathbf{x}|) = V(r)$$

Proposition 3.1

For such a potential, the corresponding force points towards or away from the origin.

Proof. We have

$$\begin{aligned} \mathbf{F} = -\boldsymbol{\nabla} V &= -\frac{dV}{dr}\boldsymbol{\nabla} r \\ &= -\frac{dV}{dr}\hat{\mathbf{x}} \end{aligned}$$

which is what we wanted. ∎

§3.1 Conservation of angular momentum

The most important fact about central forces is that they conserve angular momentum.

Definition 3.2 (Angular momentum)**.** The *angular momentum* of an object with position vector $\mathbf{x}$ is given by

$$\mathbf{L} = m\mathbf{x} \times \dot{\mathbf{x}} = \mathbf{x} \times \mathbf{p}$$

We can see that $\mathbf{L}$ is orthogonal to both position and velocity/momentum. It is important to note that $\mathbf{L}$ is defined relative to a particular origin – here we are taking the origin at $\mathbf{x} = \mathbf{0}$, the same point as the centre of the central potential.

For a general force $\mathbf{F}$, we have

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= m\frac{d}{dt}(\mathbf{x} \times \dot{\mathbf{x}}) \\ &= m(\dot{\mathbf{x}} \times \dot{\mathbf{x}} + \mathbf{x} \times \ddot{\mathbf{x}}) \\ &= \mathbf{x} \times \mathbf{F} = \mathbf{G} \end{aligned}$$

where $\mathbf{G}$ is defined to be the *torque*, the angular analogue of force.

Proposition 3.3

Central forces conserve angular momentum.

Proof. By proposition 3.1, we have $\mathbf{F} \parallel \mathbf{x}$ for such a force. Then $\frac{d\mathbf{L}}{dt} = \mathbf{x} \times \mathbf{F} = \mathbf{0}$. ∎

Furthermore, because $\mathbf{L}$ is constant, and $\mathbf{x}$ and $\dot{\mathbf{x}}$ obey

$$\mathbf{L} \cdot \mathbf{x} = 0$$
$$\mathbf{L} \cdot \dot{\mathbf{x}} = 0$$

we can see that $\mathbf{x}$ and $\dot{\mathbf{x}}$ are constrained to a plane orthogonal to $\mathbf{L}$. So, for such a force, we can reduce to a two-dimensional case.

§3.2 Polar coordinates in the plane

Consider coordinates r and θ, with

$$x = r\cos\theta$$
$$y = r\sin\theta$$

We would like to define some basis vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ in polar coordinates. In Cartesian coordinates, we have

$$\hat{\mathbf{r}} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \quad \hat{\boldsymbol{\theta}} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

and this is indeed an orthnormal basis. Of course, this basis will change as the particle moves, so care will be needed. In particular, we have

$$\frac{d\hat{\mathbf{r}}}{d\theta} = \hat{\boldsymbol{\theta}}, \quad \frac{d\hat{\boldsymbol{\theta}}}{d\theta} = -\hat{\mathbf{r}}$$

Let us now attempt to write Newton's equations in polar coordinates. We have

$$\begin{aligned}
&\mathbf{x} = r\hat{\mathbf{r}} \\
\implies &\dot{\mathbf{x}} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} \\
&\quad = \dot{r}\hat{\mathbf{r}} + r\frac{d\hat{\mathbf{r}}}{d\theta}\dot{\theta} \\
&\quad = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}} \\
\implies &\ddot{\mathbf{x}} = \ddot{r}\hat{\mathbf{r}} + 2\dot{r}\dot{\theta}\hat{\boldsymbol{\theta}} + r\ddot{\theta}\hat{\boldsymbol{\theta}} - r\dot{\theta}^2\hat{\mathbf{r}} \\
&\quad = \boxed{(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}}
\end{aligned}$$

Example 3.4 (Circular motion at constant angular speed)
We have

$$\dot{r} = 0,\ \dot{\theta} = \omega$$
$$\ddot{r} = 0,\ \ddot{\theta} = 0$$

Then

$$\ddot{\mathbf{x}} = -r\omega^2\hat{\mathbf{r}}$$

– that is, circular motion requires centripetal force towards the origin.

To write down Newton's equations for a central force, we can equate radial and angular components. We have

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0$$
$$m\left(\ddot{r} - r\dot{\theta}^2\right) = -\frac{dV}{dr}$$

Focusing for now on the easier-looking $\hat{\boldsymbol{\theta}}$ equation, we have

$$\frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0$$
$$\implies l = r^2\dot{\theta} = \text{const.}$$

In fact, this quantity l is (nearly) the magnitude of the angular momentum. We have

$$\mathbf{L} = m\mathbf{x} \times \dot{\mathbf{x}}$$
$$= mr\hat{\mathbf{r}} \times (\dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}})$$
$$= mr^2\dot{\theta}\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}}$$
$$\implies |\mathbf{L}| = ml$$

but we often call l the angular momentum regardless.

Now, let us focus on the $\hat{\mathbf{r}}$ equation. Applying the definition of l, we have

$$m\left(\ddot{r} - r\frac{l^2}{r^4}\right) = -\frac{dV}{dr}$$
$$\implies m\ddot{r} = -\frac{dV_{\text{eff}}}{dr}$$

where the *effective potential* is

$$V_{\text{eff}}(r) = V(r) + \frac{ml^2}{2r^2}$$

and now we've finally reduced to a one-dimensional problem.

§3.3 The effective potential

Consider a potential $V(r) = -\frac{k}{r}$, such as in gravity. Let us now plot V_{eff} against r – we have DIAGRAM. For small r, the $\frac{ml^2}{2r^2}$ term is most important, while for large r, the $-\frac{k}{r}$ dominates. This phenomenon is called the *centrifugal barrier* – conservation of angular momentum prevents the particle from getting too close to the origin. This is why orbits happen!

We can also see the effective potential by looking at the conserved energy. We have

$$\begin{aligned}E &= \frac{1}{2}m\dot{\mathbf{x}}\cdot\dot{\mathbf{x}} + V(r)\\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + V(r)\\ &= \frac{1}{2}m\dot{r}^2 + V(r) + \frac{ml^2}{2r^2}\\ &= \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r)\end{aligned}$$

and here it's clear that the centrifugal barrier is just a product of the angular kinetic energy.

DIAGRAM: horizontal energy lines on the potential graph correspond to hyperbolic, parabolic, elliptical and circular orbits. Closest point for elliptical periapsis/perihelion, furthest apoapsis/aphelion.

Let us quickly consider a world in which the potential takes the form $-kr^{-n}$, $n > 2$. Then

DIAGRAM: potential graph, inverted.

Now we don't have any stable orbits! In fact, gravity in d spatial dimensions has $V \propto \frac{1}{r^{d-2}}$ – so circular orbits are only stable in dimensions $d < 4$.

Let us now solve the equations of motion.

§3.4 The orbit equation

The cleanest solution starts with a clever change of variables, to $u = \frac{1}{r}$. We now want to find the orbit equation for $u(\theta)$ (instead of $r(t)$). We have

$$\begin{aligned}\frac{dr}{dt} &= \frac{dr}{d\theta}\dot{\theta} = \frac{dr}{d\theta}\frac{l}{r^2} = -l\frac{du}{d\theta}\\ \frac{d^2r}{dt^2} &= \frac{d}{dt}\left(-l\frac{du}{d\theta}\right) = -l\frac{d^2u}{d\theta^2}\dot{\theta} = -l^2u^2\frac{d^2u}{d\theta^2}\end{aligned}$$

We now have

$$\begin{aligned}m\ddot{r} - \frac{ml^2}{r^3} &= F(r)\\ \implies -l^2mu^2\frac{d^2u}{d\theta^2} - mu^3l^2 &= F\left(\frac{1}{u}\right)\\ \implies \frac{d^2u}{d\theta^2} + u &= -\frac{1}{ml^2u^2}F\left(\frac{1}{u}\right)\end{aligned}$$

Of course, in our case, we have $V = -\frac{km}{r}$ (where $k = GM$). This gives

$$\begin{aligned}\frac{d^2u}{d\theta^2} + u &= \frac{k}{l^2}\\ \implies u &= A\cos(\theta - \theta_0) + \frac{k}{l^2}\end{aligned}$$

We can see that u is largest at $\theta = \theta_0$, so r is smallest there – this is the periapsis. We will choose our axes such that $\theta_0 = 0$. We obtain

$$\boxed{r = \frac{r_0}{e\cos\theta + 1}}$$

where $r_0 = \frac{l^2}{k}$ is a fixed constant, and e is a constant of integration. This is a conic section!

The constant of integration e is called the *eccentricity*. We have that

- $e = 0$ is a *circular orbit*;
- $0 < e < 1$ is a *elliptical orbit*;
- $e = 1$ is a *parabolic* orbit;
- $e > 1$ is a *hyperbolic* orbit.

Let us consider the elliptical case. We have

$$\begin{aligned}
&(r_0 - re\cos\theta)^2 = r^2 \\
&\implies (r_0 - ex)^2 = x^2 + y^2 \\
\implies &\frac{(x - x_c)^2}{a^2} + \frac{y^2}{b^2} = 1
\end{aligned}$$

where $r = 0$ is the *focus*, not the centre. The semi-axes a and b are given in terms of e and r_0, as is x_c.

Planets in the solar system have small e. The most elliptical orbit, Mercury, has $e \approx 0.2$. Halley's comet, on the other hand, has $e \approx 0.97$.

Let us consider the parabolic case. We have

$$\begin{aligned}
&r_0^2 - 2r_0 x + x^2 = x^2 + y^2 \\
&\implies r_0^2 - 2r_0 x = y^2
\end{aligned}$$

In the hyperbolic case, we see that $r \to \infty$ at $\cos\theta = -\frac{1}{e}$ (in particular, $\theta > \pi/2$, so with our convention of $\theta_0 = 0$, it goes to infinity on the left).

One important result allows us to evaluate the energy of a particular solution. We have

$$\begin{aligned}
E &= \frac{1}{2}m\dot{r}^2 + \frac{ml^2}{2r^2} - \frac{km}{r} \\
&= \cdots \\
&= \frac{mk^2}{2l^2}(e^2 - 1)
\end{aligned}$$

§3.5 Kepler's laws

Historically, these laws were derived before the orbit equation was ever solved. Kepler derived them empirically, while Newton justified them mathematically.

Proposition 3.5 (Kepler's first law)

Each planet moves in an ellipse, with the sun at one focus.

Proposition 3.6 (Kepler's second law)

The line between a planet and the sun sweeps out equal areas in equal times.

DIAGRAM

Proof. We have

$$\begin{aligned}\delta A &= \frac{1}{2}r^2\delta\theta \\ \implies \dot{A} &= \frac{1}{2}r^2\dot{\theta} \\ &= \frac{l}{2}\end{aligned}$$

which is constant by conservation of angular momentum. ■

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Proposition 3.7 (Kepler's third law)
The orbital period T is proportional to the radius $R^{3/2}$.

Dimensional analysis. It's natural to first show this using dimensional analysis. The only parameter in Newton's equation is $k = GM$ (since m cancels). Hence, the period is

$$\begin{aligned}T &= cR^A k^B \\ \implies \mathrm{L} &= \mathrm{L}^A \mathrm{L}^{3B}\mathrm{T}^{-2B}\end{aligned}$$

which gives $B = -\frac{1}{2}$, $A = \frac{3}{2}$, giving $T = c\frac{R^{3/2}}{k^{1/2}}$ as required.

We should remark that, since there is no canonical 'radius' for an ellipse, this c depends on whether we pick the semi-major axis, semi-minor axis or anything in between. ■

Direct. It is in fact true that K3 only holds for inverse-square law orbits, which is a subtlety we would not notice doing dimensional analysis. So let us derive the law directly. Recall that $\frac{dA}{dt} = \frac{l}{2}$. Then

$$\begin{aligned}T &= \int_0^T dt = \int_0^A \frac{2}{l}\,dA \\ &= \frac{2\pi ab}{l} \\ &= \frac{2\pi}{l}\cdot\frac{r_0^2}{(1-e^2)^{3/2}}\end{aligned}$$

Let us recall also that $l^2 = kr_0$, so we have

$$\begin{aligned}T &= \frac{2\pi}{\sqrt{GM}}\left(\frac{r_0}{1-e^2}\right)^{3/2} \\ &= \frac{2\pi}{\sqrt{GM}}R_{\text{avg}}^{3/2}\end{aligned}$$

because

$$\frac{r_0}{1-e^2} = \frac{1}{2}\left(\frac{r_0}{1+e} + \frac{r_0}{1-e}\right)$$

which are the minimum and maximum radii respectively. This concludes the proof. ■

§3.6 Repulsive potentials and scattering

Given a central potential $V(r)$ satisfying $V \to 0$ as $r \to \infty$, one can perform *scattering experiments* – shoot a particle in from large r and see how it flies out again. We have an important new concept.

Definition 3.8 (Impact parameter)**.** The *impact parameter* for a scattering experiment, b, is the closest the particle would have come to the target in the absence of forces.

Proposition 3.9
The impact parameter b satisfies

$$l = bv$$

where l is the (specific) angular momentum, and v is the particle's initial velocity.

Proof. If we imagine the particle proceeding along the path without forces, it will have velocity v at the point of closest approach, in particular b. At this point, the vector $\mathbf{v}$ is orthogonal to the radial vector from the central object to the particle, so $|\mathbf{v} \times \mathbf{r}| = |\mathbf{v}||\mathbf{r}| = vb$. Since angular momentum is conserved, this must hold for the particle anywhere along its path, even when forces are applied. ∎

Let us consider the Rutherford scattering experiment, which showed the existence of the nucleus. For two positively charged particles, we have a potential of

$$V = \frac{\kappa}{r}$$

where κ is $\frac{qQ}{4\pi\varepsilon_0}$ (this is just Coulomb's law).

We may reuse results from the Kepler problem, replacing all instances of $-km$ with κ. In particular, the orbits are still

$$r = \frac{r_0}{\tilde{e}\cos\theta - 1}$$

with $r_0 = \frac{l^2 m}{\kappa}$ (note we've changed some signs but it is in fact the same equation). We want to find the scattering angle φ between the particle's incoming and outgoing paths.

First, note the incoming and outgoing impact parameters are equal, since angular momentum is conserved and, at a given distance, the speed is the same by symmetry or the conservation of energy. Looking at the orbit equation, we can clearly see that r goes to infinity when the denominator is zero, i.e. $\tilde{e}\cos\theta = 1$. Let us consider energies. We have

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{\kappa^2}{2l^2 m}\left(\tilde{e}^2 - 1\right)\\ &= \frac{\kappa^2}{2m(bv)^2}\left(\frac{1}{\cos^2\theta} - 1\right)\\ &= \frac{\kappa^2}{2mb^2v^2}\tan^2\theta\end{aligned}$$

Also, we can see (DIAGRAM!!!!) that the angle of deviation of the particle's path, φ, satisfies $\varphi + 2\theta = \pi$, so

$$\tan\theta = \tan\left(\frac{\pi - \varphi}{2}\right) = \frac{1}{\tan\frac{\varphi}{2}}$$

Hence

$$\begin{aligned} \frac{1}{2}mv^2 &= \frac{\kappa^2}{2mb^2v^2} \cdot \frac{1}{\tan^2 \frac{\varphi}{2}} \\ \implies \varphi &= 2\arctan\left(\frac{\kappa}{bmv^2}\right) \end{aligned}$$

We see that a small b leads to a large scattering angle, which allows scattering experiments to probe very small distances.

§4 Systems of particles

Let us consider now systems of N particles, labelled with $i = 1, \dots, n$. Each particle has position $\mathbf{x}_i$, mass m_i and momentum $\mathbf{p}_i = m_i\dot{\mathbf{x}}_i$. And of course each particle obeys Newton's law.

The force $\mathbf{F}_i$ on the i^{th} particle can be either *external* or due to the other particles:

$$\mathbf{F}_i = \mathbf{F}_i^{\text{ext}} + \sum_{i \neq j} \mathbf{F}_{ij}$$

where $\mathbf{F}_{ij}$ is the force on the i^{th} particle due to the j^{th} particle.

We have the following important result:

Theorem 4.1 (Newton's third law)
For two particles with labels i, j,

$$\mathbf{F}_{ij} = -\mathbf{F}_{ji}$$

– or, in words, 'every action has an equal and opposite reaction.'

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Let us consider some consequence of Newton's third law for systems of particles.

§4.1 Centre of mass

The total mass is $M = \sum_{i=1}^{N} M_i$. We define the *centre of mass*

$$\mathbf{R} = \frac{1}{M}\sum_{i=1}^{N} M_i\mathbf{x}_i$$

so the total momentum is

$$\begin{aligned} \mathbf{P} &= \sum_{i=1}^{N} \mathbf{p}_i \\ &= \sum_i m_i\dot{\mathbf{x}}_i \\ &= m\dot{\mathbf{R}} \end{aligned}$$

so the centre of mass behaves like a single particle of mass M. Furthermore,

$$\begin{aligned}\dot{\mathbf{P}} &= \sum_i \dot{\mathbf{p}}_i \\ &= \sum_i (\mathbf{F}_i^{\text{ext}} + \sum_{j \neq i} \mathbf{F}_{ij}) \\ &= \sum_i \mathbf{F}_i^{\text{ext}} + \sum_{i<j} \mathbf{F}_{ij} + \mathbf{F}_{ji} \\ &= \mathbf{F}, \text{ the total external force}\end{aligned}$$

where the second term in the penultimate line is zero by Newton's third law. We conclude that $M\ddot{\mathbf{R}} = \mathbf{F}$, the total external force – we are allowed to view composite objects like point particles. In particular, $\mathbf{F} = \mathbf{0} \implies \dot{\mathbf{p}} = \mathbf{0}$ – total momentum is conserved in the absence of external forces.

We would like to show the same result for total *angular* momentum. Let us work relative to an arbitrary fixed point $\mathbf{a}$: we have

$$\begin{aligned} \mathbf{L} &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \mathbf{p}_i \\ \implies \dot{\mathbf{L}} &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \dot{\mathbf{p}}_i \\ &= \sum_i (\mathbf{x}_i - \mathbf{a}) \times \left(\mathbf{F}_i^{\text{ext}} + \sum_{j \neq i} \mathbf{F}_{ij} \right) \\ &= \mathbf{G} + \sum_{i<j} \left((\mathbf{x}_i - \mathbf{a}) \times \mathbf{F}_{ij} - (\mathbf{x}_j - \mathbf{a}) \times \mathbf{F}_{ij} \right) \\ &= \mathbf{G} + \sum_{i<j} (\mathbf{x}_i - \mathbf{x}_j) \times \mathbf{F}_{ij} \qquad (\dagger) \end{aligned}$$

Here the total torque

$$\mathbf{G} = \sum_i (\mathbf{x}_i - \mathbf{a}) \times \mathbf{F}_i^{\text{ext}}$$

but the vanishing of the second term in (†) is not a given.

Let us consider the special case where the interaction forces come from a potential that depends on the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$, that is

$$\mathbf{F}_{ij} = -\boldsymbol{\nabla}_i V(|\mathbf{x}_i - \mathbf{x}_j|)$$

where we are taking the gradient with respect to $\mathbf{x}_i$ This is

$$\mathbf{F}_{ij} = -V'(|\mathbf{x}_i - \mathbf{x}_j|) \frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|}$$

from which it clearly follows that $(\mathbf{x}_i - \mathbf{x}_j) \times \mathbf{F}_{ij} = \mathbf{0}$. In this case, $\dot{\mathbf{L}} = \mathbf{G}$.

In fact, there are tools that allow us to show that the second term in (†) will vanish for all known forces, but we cannot do this right now. In general, we conclude $\dot{\mathbf{L}} = \mathbf{G}$ – an object cannot spin itself.

It's natural to take $\mathbf{a}$, our 'centre', to be the centre of mass. However, in general, the centre of mass will change as a function of time, which will require us to redo the derivation above. But all will be well.

The total *energy* also splits up nicely: we have

$$\mathbf{x}_i = \mathbf{R} + \mathbf{y}_i$$

where $\mathbf{y}_i$ is the position of the i^{th} particle relative to the centre of mass. But, since

$$M\mathbf{R} = \sum_i m_i \mathbf{x}_i \tag{1}$$

we conclude that $\sum_i m_i \mathbf{y}_i = \mathbf{0}$ for all time. Then

$$\begin{aligned} T &= \sum_i \frac{1}{2} m_i \dot{\mathbf{x}}_i \cdot \dot{\mathbf{x}}_i \qquad (2) \\ &= \sum_i \frac{1}{2} m_i \left(\dot{\mathbf{R}}^2 + \dot{\mathbf{y}}_i^2 + 2\dot{\mathbf{R}}\dot{\mathbf{y}}_i \right) \qquad (3) \\ &= \frac{1}{2} M \dot{\mathbf{R}}^2 + \sum_i \frac{1}{2} m_i \dot{\mathbf{y}}_i^2 \qquad (4) \end{aligned}$$

so we also have notions of the internal kinetic energy and the kinetic energy which can be in some sense attributed to the centre of mass.

Let us now consider potential energy, and work as usual under the assumption that all forces are conservative. We have

$$\begin{aligned} \mathbf{F}_i^{\text{ext}} &= -\boldsymbol{\nabla}_i V_i(\mathbf{x}_i) \\ \mathbf{F}_{ij} &= -\boldsymbol{\nabla}_i V_{ij}(|\mathbf{x}_i - \mathbf{x}_j|) \end{aligned}$$

where $V_{ij} = V_{ji}$ to respect N3. We now have a conserved energy.

Proposition 4.2
The total energy

$$E = T + \sum_i V_i(\mathbf{x}_i) + \sum_{i<j} V_{ij}(|\mathbf{x}_i - \mathbf{x}_j|)$$

is conserved. *Note:* each potential term only appears once!

Proof. We have

$$\begin{aligned} \frac{dE}{dt} &= \sum_i m_i \dot{\mathbf{x}}_i \cdot \ddot{\mathbf{x}}_i + \sum_i \boldsymbol{\nabla}_i V_i \cdot \dot{\mathbf{x}}_i + \sum_{i<j} \left(\boldsymbol{\nabla}_i V_{ij} \cdot \dot{\mathbf{x}}_i + \boldsymbol{\nabla}_j V_{ij} \cdot \dot{\mathbf{x}}_j \right) \\ &= \sum_i m_i \dot{\mathbf{x}}_i \cdot \ddot{\mathbf{x}}_i - \sum_i \mathbf{F}_i^{\text{ext}} \cdot \dot{\mathbf{x}}_i - \sum_{i<j} \left(\mathbf{F}_{ij} \cdot \dot{\mathbf{x}}_i + \mathbf{F}_{ji} \cdot \dot{\mathbf{x}}_j \right) \\ &= \sum_i \dot{\mathbf{x}}_i \cdot \left(m\ddot{\mathbf{x}}_i - \mathbf{F}_i^{\text{ext}} - \sum_{i \neq j} \mathbf{F}_{ij} \right) \\ &= 0 \end{aligned}$$

by Newton's second law. ■

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§4.2 The two-body problem

The two-body problem is an important special system with two particles and no external forces, for example, the Earth and Moon.

Let $\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$ be the relative separation of the particles. The centre of mass, $\mathbf{R}$, is given by $M\mathbf{R} = m_1\mathbf{x}_1 + m_2\mathbf{x}_2$, so that

$$\mathbf{x}_1 = \mathbf{R} + \underbrace{\frac{m_2}{M}\mathbf{r}}_{\mathbf{y}_1}$$

$$\mathbf{x}_2 = \mathbf{R} - \underbrace{\frac{m_1}{M}\mathbf{r}}_{\mathbf{y}_2}$$

so our kinetic energy is

$$\begin{aligned} T &= \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}m_1\frac{m_2^2}{M^2}\dot{\mathbf{r}}^2 + \frac{1}{2}m_2\frac{m_1^2}{M^2}\dot{\mathbf{r}}^2 \\ &= \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}\mu\dot{\mathbf{r}}^2 \end{aligned}$$

where $\mu = \frac{m_1 m_2}{m_1+m_2}$ is the *reduced mass*. With no external forces, we know $\ddot{\mathbf{R}} = 0$ – the centre of mass doesn't accelerate. Let us consider

$$\begin{aligned} \mu\ddot{\mathbf{r}} &= \mu(\ddot{\mathbf{x}}_1 - \ddot{\mathbf{x}}_2) \\ &= \mu\left(\frac{\mathbf{F}_{12}}{m_1} - \frac{\mathbf{F}_{21}}{m_2}\right) \\ &= \mu\left(\frac{1}{m_1} + \frac{1}{m_2}\right)\mathbf{F}_{12} = \mathbf{F}_{12} \end{aligned}$$

so the relative separation behaves as in a single particle problem.

Furthermore, if $m_1 \gg m_2$, then $\mu = \frac{m_1 m_2}{m_1+0} = m_2$, so the reduced mass is just the smaller mass – the heavy object is essentially still and the light object orbits around it.

In general, for the number of particles $N > 2$, one cannot solve. However, if $N \gg 1$, we can use new models – statistical physics.

§4.3 Rocket equation and variable mass

The relation $\dot{\mathbf{P}} = \mathbf{F}^{\text{ext}}$ is very useful when the internal forces are very complicated. Consider, for example, a rocket that ejects fuel with speed u relative to the rocket. The process of ejection may be complicated, but total momentum must obey

$$\dot{P} = \frac{P(t+\delta t) - P(t)}{\delta t} = F^{\text{ext}}$$

where everything is now in the z-direction. We have

$$P(t) = m(t)v(t)$$

$$\begin{aligned} P(t+\delta t) &= m(t+\delta t)v(t+\delta t) + (m(t) - m(t+\delta t))(v(t) - u) \\ &= m(t)v(t) + [m'(t)v(t) + m(t)v'(t)]\delta t - m'(t)(v(t) - u)\delta t \end{aligned}$$

$$\implies \boxed{m\dot{v} + u\dot{m} = F^{\text{ext}}}$$

This is called the *Tsiolkovsky rocket equation*. We might also write

$$m\dot{v} = F^{\text{ext}} - u\dot{m}$$

and we can identify $u\dot{m}$ as the *thrust force*.

§5 Rigid bodies

We can think of rigid body problems as N-body problems in which the distances between the particles are fixed.

The only motions a rigid body can undergo are translations of the centre of mass, and rotations.

§5.1 Angular velocity

In three dimensions, rotations are described by a vector called the *angular velocity* $\boldsymbol{\omega}$.

Definition 5.1. The *angular velocity* $\boldsymbol{\omega}$ of an object is such that

$$\boldsymbol{\omega} = \omega \hat{\mathbf{n}}$$

such that $\hat{\mathbf{n}}$ points along the axis of rotation and $\omega = |\boldsymbol{\omega}| = \dot{\theta}$ is the angular speed. The direction that $\boldsymbol{\omega}$ points is given by the right-hand grip rule, where the thumb gives the direction of $\boldsymbol{\omega}$ and the fingers give the direction of rotation.

Proposition 5.2

We have

$$\dot{\mathbf{x}} = \boldsymbol{\omega} \times \mathbf{x}$$

DIAGRAM!!!

In addition to the angular velocity, a rotation must specify a point through which the axis of rotation passes – in proposition 5.2, the choice of point is implicit in the choice of origin for the vector $\mathbf{x}$.

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§5.2 Moment of inertia

The rotation of a particle involves kinetic energy. We have

$$\begin{aligned} T &= \frac{1}{2} m \dot{\mathbf{x}}^2 \\ &= \frac{1}{2} m (\boldsymbol{\omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega} \times \mathbf{x}) \\ &= \frac{1}{2} m \omega^2 d^2 \end{aligned}$$

where $d = |\hat{\mathbf{n}} \times \mathbf{x}|$ is the perpendicular distance of the particle from the axis of rotation.

In a rigid body, all particles rotate with the same angular velocity (in order to keep the distances between the particles fixed) – we have

$$\begin{aligned} \frac{d}{dt} |\mathbf{x}_i - \mathbf{x}_j|^2 &= 2(\dot{\mathbf{x}}_i - \dot{\mathbf{x}}_j) \cdot (\mathbf{x}_i - \mathbf{x}_j) \\ &= 2 \left(\boldsymbol{\omega} \times (\mathbf{x}_i - \mathbf{x}_j)\right) \cdot (\mathbf{x}_i - \mathbf{x}_j) \\ &= 0 \end{aligned}$$

Definition 5.3 (Moment of inertia)**.** The *moment of inertia* of a rigid body with particles labelled by i is

$$I = \sum_i m_i d_i^2$$

where $d_i = |\hat{\mathbf{n}} \times \mathbf{x}_i|$ is the distance of the i^{th} particle from the axis of rotation.

The motivation of this definition is that the kinetic energy of a rigid body is then

$$\begin{aligned} T &= \frac{1}{2}\sum_i m_i \dot{\mathbf{x}}_i \cdot \dot{\mathbf{x}}_i \\ &= \frac{1}{2} I\omega^2 \end{aligned}$$

> **Remark.** An object's moment of inertia is not an intrinsic property – it depends on the axis around which it is rotating.

The angular momentum of a rotating rigid body is then

$$\begin{aligned} \mathbf{L} &= \sum_i m_i \mathbf{x}_i \times \dot{\mathbf{x}}_i \\ &= \sum_i m_i \mathbf{x}_i \times (\boldsymbol{\omega} \times \mathbf{x}_i) \end{aligned}$$

We will for now only consider the component of $\mathbf{L}$ along the direction of $\boldsymbol{\omega}$, which is

$$\begin{aligned} L &= \mathbf{L} \cdot \hat{\mathbf{n}} \\ &= \omega \sum_i m_i \left(\mathbf{x}_i \times (\hat{\mathbf{n}} \times \mathbf{x}_i)\right) \\ &= \omega \sum_i m_i (\hat{\mathbf{n}} \times \mathbf{x}_i) \cdot (\hat{\mathbf{n}} \times \mathbf{x}_i) \\ &= \omega \sum_i m_i d_i^2 \\ &= \omega I \end{aligned}$$

and once again I plays the role of mass.

Recall that the application of a torque causes a change in the angular momentum – $\dot{\mathbf{L}} = \mathbf{G}$. If the torque is also around the axis of rotation, then $\mathbf{G} = G\hat{\mathbf{n}}$, and

$$\begin{aligned} \mathbf{G} &= \dot{\mathbf{L}} \\ \implies \hat{\mathbf{n}} \cdot \mathbf{G} &= \hat{\mathbf{n}} \cdot \dot{\mathbf{L}} \\ \implies G &= I\dot{\omega} \end{aligned}$$

Typically we calculate the moment of inertia by taking $N \to \infty$ and turning our sums into integrals – we get

$$\sum_i m_i f(\mathbf{x}_i) \approx \int d^3x\, \rho(\mathbf{x}) f(\mathbf{x})$$

where $\rho(\mathbf{x})$ is the *density*. We will also typically consider a uniform density, so $\rho(\mathbf{x}) = \rho_0$ a constant. Hence we have

$$\begin{aligned} M &= \rho \int d^3x \\ I &= \rho \int d^3x\, x_\perp^2 \end{aligned}$$

Example

- *Hoop*. Consider a circular hoop rotating around a perpendicular axis through its centre. DIAGRAM We have

$$M = 2\pi a\rho$$
$$I = 2\pi a^3\rho = Ma^2$$

- *Rod*. Consider a rod rotating around one of its endpoints. We have

$$M = l\rho$$
$$I = \rho\int_0^l dx\, x^2 = \frac{1}{3}\rho l^3 = \frac{1}{3}Ml^2$$

- *Disc*. Consider a circular disc rotating around a perpendicular axis through its centre. We have

$$M = \pi a^2\rho$$
$$I = 2\pi\rho\int_0^a r\,dr\cdot r^2 = 2\pi\rho\frac{a^4}{4} = \frac{1}{2}Ma^2$$

where the $r\,dr$ appears from the Jacobian.

- *Sphere* Consider a sphere rotating about any axis through its centre. We have

$$M = \frac{4}{3}\pi a^3\rho$$
$$I = 2\pi\rho\int_0^a\int_0^\pi r^2\sin\theta\,drd\theta\, r^2\sin^2\theta$$
$$= 2\pi\rho\frac{a^5}{5}\cdot\frac{4}{3} = \frac{2}{5}Ma^2$$

§5.3 Perpendicular axis theorem

Proposition 5.4 (Perpendicular axis theorem)
Consider any 2D body (a 'lamina'), lying in the x-y plane, and let I_x, I_y and I_z be moments of inertia around those axes. Then

$$I_z = I_x + I_y$$

Proof. Clearly

$$I_x = \int d^3x\,\rho y^2$$
$$I_y = \int d^3x\,\rho x^2$$
$$I_z = \int d^3x\,\rho(x^2+y^2)$$

from which the result is immediate. ∎

§5.4 Parallel axis theorem

Proposition 5.5 (Parallel axis theorem)

Consider an axis through a body's centre of mass and an axis parallel to the above, separated by a distance h. Let $I_{\text{c.o.m.}}$ and I be the moments of inertia of rotation about these axes respectively. Then

$$I = I_{\text{c.o.m.}} + Mh^2$$

Proof. Set $\mathbf{x}_i = \mathbf{R} + \mathbf{y}_i$. Then, recalling a unit vector along the rotation axis is given by $\hat{\mathbf{n}}$,

$$\begin{aligned}
I &= \sum_i m_i d^2 = \sum_i m_i (\hat{\mathbf{n}} \times \mathbf{x}_i) \cdot (\hat{\mathbf{n}} \times \mathbf{x}_i) \\
&= \sum_i m_i (\hat{\mathbf{n}} \times (\mathbf{R} + \mathbf{y}_i)) \cdot (\hat{\mathbf{n}} \times (\mathbf{R} + \mathbf{y}_i)) \\
&= \sum_i m_i \left((\hat{\mathbf{n}} \times \mathbf{R}) \cdot (\hat{\mathbf{n}} \times \mathbf{R}) + (\hat{\mathbf{n}} \times \mathbf{y}_i) \cdot (\hat{\mathbf{n}} \times \mathbf{y}_i) + 2(\hat{\mathbf{n}} \times \mathbf{R}) \cdot (\hat{\mathbf{n}} \times \mathbf{y}_i) \right) \\
&= Mh^2 + I_{\text{c.o.m.}} + 0
\end{aligned}$$

where the last term vanishes by linearity of the cross product and the fact that $\sum_i m_i \mathbf{y}_i = 0$. ■

Remark. The theorem shows that, for any given axis of rotation, the moment of inertia is minimized when rotating around an axis through the object's centre of mass.

§5.5 Motion of rigid bodies

We've established that rigid bodies should only be able to move and rotate. We claim we can model particles in a rigid body by

$$\mathbf{x}_i = \mathbf{R}(t) + \mathbf{y}_i$$

where $\mathbf{y}_i$ satisfies $\dot{\mathbf{y}}_i = \boldsymbol{\omega} \times \mathbf{y}_i$ – in other words, each $\mathbf{y}_i$ is rotating only with the same angular velocity.

Let us now consider the energy. Starting from our previous result,

$$\begin{aligned}
T &= \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} \sum_i m_i \dot{\mathbf{y}}_i^2 \\
&= \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} \sum_i m_i (\boldsymbol{\omega} \times \mathbf{y}_i)^2 \\
&= \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} I_{\text{c.o.m.}} \omega^2
\end{aligned}$$

where these terms are the *translational* and *rotational* kinetic energies respectively.

What about the potential energy? We have

$$V = \sum_i V_i(\mathbf{x}_i) + \sum_{i<j} V_{ij}(|\mathbf{x}_i - \mathbf{x}_j|)$$

The second term is constant, since the distances between particles do not change – hence it drops out upon taking grad.

What about the first term? Let's consider a nice potential such as a cosntant gravitational field,

$$\begin{aligned} V_i(\mathbf{x}_i) &= m_i g z_i \\ \implies V = \sum_i V_i &= g \sum_i m_i z_i \\ &= gMR_z \end{aligned}$$

where R_z is the z-component of the centre of mass – in other words, a rigid body behaves just like a point particle under the influence of gravity.

Example (Pendulum)

Consider a rigid rod pendulum with mass M and length L, fixed at some point and free to rotate under the influence of gravity. DIAGRAM.

Our approach will be to find the energy and then set $\frac{dE}{dt} = 0$. We will find E by two methods.

1. Observe the pivot is fixed, so there is only rotational kinetic energy. Hence

$$T = \frac{1}{2} I \dot{\theta}^2$$

where $\dot{\theta} = \omega$ and $I = \frac{1}{3}ML^2$.

2. Consider the centre of mass, which moves with velocity $\frac{l\dot{\theta}}{2}$. Then

$$T = \frac{1}{2} M \left(\frac{l\dot{\theta}}{2} \right)^2 + \frac{1}{2} I_{\text{c.o.m.}} \dot{\theta}^2$$

where $I = \frac{1}{3}ML^2 - M\left(\frac{L}{2}\right)^2$ by the parallel axis theorem. Why is the angular velocity around the centre of mass the same as the centre of mass around the pivot? DIAGRAM! We obtain

$$T = \frac{1}{2} I \dot{\theta}^2$$

as expected.

Now

$$E = \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} MgL \cos\theta$$

as can be seen by considering a diagram. Hence, after differentiating and cancelling $\dot{\theta}$, we obtain

$$I\ddot{\theta} = -\frac{1}{2} MgL \sin\theta$$

Example (Rolling ball)

No-slip rolling occurs when the friction between the ball and the ground is so strong that the point of contact is not moving. DIAGRAM!!! In particular, we have the result $v = a\dot{\theta}$, where a is the radius of the ball and v its velocity.

The kinetic energy is

$$\begin{aligned} T &= \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}v^2\left(M + \frac{I}{a^2}\right) \end{aligned}$$

Importantly, normally including friction in our models will lead to energy not being conserved – but, because there is no relative velocity at the point of contact, no work is done by the friction, and so energy is conserved.

Let us, then, put our ball on a slope of incline α and have it roll down. We have

$$E = \frac{1}{2}\left(M + \frac{I}{a^2}\right)\dot{x}^2 - Mgx\sin\alpha$$

so, once again differentiating and dividing by $\dot{x}$, we have

$$\left(M + \frac{I}{a^2}\right)\ddot{x} = Mg\sin\alpha$$

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Let us 'rigorously' show that energy is conserved. We have

$$\begin{aligned} E &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}I\dot{\theta}^2 \\ \implies \frac{dE}{dt} &= M\dot{x}\ddot{x} + I\dot{\theta}\ddot{\theta} \\ &= -F\dot{x} + aF\dot{\theta} \\ &= F(-\dot{x} + a\dot{\theta}) = 0 \end{aligned}$$

where $\dot{x} + a\dot{\theta}$ is exactly the fact that the ball is rolling.

§5.6 Normal forces

Objects don't fall through tables, etc., because the table exerts a *normal force* on the object that pushes it away. Much like friction, this force is a macroscopic approximation to microscopic interactions such as electrostatic repulsion and the Pauli exclusion principle.

DIAGRAM – object on table with gravitational and normal forces.

The normal force is most important when we are on a slope.

DIAGRAM: slope θ above horizontal, block of mass mg, forces resolved, dry friction F_F also labelled.

Normal force doesn't prevent the object from sliding – that is due to the *dry friction* force, F_F. Typically, once the object starts moving, we have

$$F_F = \mu F_N$$

where μ, the coefficient of friction, is a dimensionless constant. Note that the friction is independent of speed, but always opposes the direction of motion.

Normal forces also produce elastic bounces:

DIAGRAM

The normal force on impact is in the direction normal to the surface at the point of impact. Let us show that the angle of incidence equals the angle of reflection.

First, assume that total energy is conserved – the collision is *elastic*. By momentum conservation (in the y-direction), we have

$$p_y = q_y + Q_y$$

where Q_y is the momentum of the Earth/wall imparted by the collision. Conservation of energy (in the y-direction) gives

$$\begin{aligned} \frac{p_y^2}{2m} &= \frac{q_y^2}{2m} + \frac{Q_y^2}{2M} \\ \implies \frac{(q_y + Q_y)^2}{2m} &= \frac{q_y^2}{2m} + \frac{Q_y^2}{2M} \end{aligned}$$

If $M \gg m$, $\frac{Q_y^2}{2M} \ll \frac{Q_y^2}{2m}$, so we can drop the $\frac{Q_y^2}{2M}$. Hence

$$\begin{aligned} Q_y &= -2q_y \\ \implies p_y &= -q_y \end{aligned}$$

so $\beta = \alpha$ as we expect.

> **Remark.** This problem involves an almost instantaneous change in momentum – this is called an *impulse*.

§6 Rotating reference frames

Rotating reference frames are important examples of non-inertial frames.

§6.1 Newton's equations in a rotating reference frame

Theorem 6.1 (Newton's equations in a rotating reference frame)
In a reference frame with angular velocity $\boldsymbol{\omega}$,

$$\left(\frac{d^2\mathbf{x}}{dt^2}\right)_{S'} = \frac{\mathbf{F}}{m} - \dot{\boldsymbol{\omega}} \times \mathbf{x} - 2\boldsymbol{\omega} \times \left(\frac{d\mathbf{x}}{dt}\right)_{S'} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{x})$$

where the notation indicates the component-wise derivatives, for components with respect to a rotating basis.

Proof. Let us consider an inertial frame S with Cartesian axes $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and a rotating frame S' with axes $\mathbf{e}_1'$, $\mathbf{e}_2'$, $\mathbf{e}_3'$.

From the perspective of the inertial frame, the $\mathbf{e}_i'$ axes rotate with angular velocity $\boldsymbol{\omega}$ – that is,

$$\dot{\mathbf{e}}_i' = \boldsymbol{\omega} \times \mathbf{e}_i'$$

In the two frames, the position of a particle is, respectively,

$$\begin{aligned}
\mathbf{x} &= x_i\mathbf{e}_i = x_i'\mathbf{e}_i' \\
\implies \dot{\mathbf{x}} &= \dot{x}_i'\mathbf{e}_i' + x_i'\dot{\mathbf{e}}_i' \\
&= \dot{x}_i'\mathbf{e}_i' + x_i'\boldsymbol{\omega}\times\mathbf{e}_i' \\
&= \dot{x}_i'\mathbf{e}_i' + \omega\times\mathbf{x} \\
&= \left(\frac{d\mathbf{x}}{dt}\right)_{S'} + \boldsymbol{\omega}\times\mathbf{x} \\
\implies \left(\frac{d\mathbf{x}}{dt}\right)_S &= \left(\frac{d\mathbf{x}}{dt}\right)_{S'} + \boldsymbol{\omega}\times\mathbf{x}
\end{aligned}$$

where this notation indicates coordinate-wise differentiation in a given frame. This makes sense – the difference, $\boldsymbol{\omega}\times\mathbf{x}$, is the relative velocity of the frames.

Now for the acceleration!

$$\begin{aligned}
\ddot{\mathbf{x}} &= \ddot{x}_i'\mathbf{e}_i' + \dot{x}_i'\dot{\mathbf{e}}_i' + \dot{\boldsymbol{\omega}}\times\mathbf{x} + \boldsymbol{\omega}\times\dot{\mathbf{x}} \\
\implies \left(\frac{d^2\mathbf{x}}{dt^2}\right)_S &= \left(\frac{d^2\mathbf{x}}{dt^2}\right)_{S'} + \boldsymbol{\omega}\times\left(\frac{d\mathbf{x}}{dt}\right)_{S'} + \dot{\boldsymbol{\omega}}\times\mathbf{x} + \boldsymbol{\omega}\times\left(\left(\frac{d\mathbf{x}}{dt}\right)_{S'} + \boldsymbol{\omega}\times\mathbf{x}\right) \\
&= \left(\frac{d^2\mathbf{x}}{dt^2}\right)_{S'} + \dot{\boldsymbol{\omega}}\times\mathbf{x} + 2\boldsymbol{\omega}\times\left(\frac{d\mathbf{x}}{dt}\right)_{S'} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{x})
\end{aligned}$$

The result follows. ■

We can see that the change to a rotating reference frame manifests itself as three so-called *fictitious forces*:

- $-\dot{\boldsymbol{\omega}}\times\mathbf{x}$ – the *Euler force*;
- $-2\boldsymbol{\omega}\times\left(\frac{d\mathbf{x}}{dt}\right)_{S'}$ – the *Coriolis force*;
- $-\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{x})$ – the *centrifugal force*.

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One important example of a rotating frame is any point on Earth. We have

$$\begin{aligned}
\omega_{\text{rot}} &= \frac{2\pi}{1\text{day}} \approx 7\times10^{-5}\text{s}^{-1} \\
R_{\text{earth}} &\approx 6\times10^{3}\text{km}
\end{aligned}$$

We will neglect the small 'wobble' of Earth – hence $\dot{\boldsymbol{\omega}} = \mathbf{0}$, there is no Euler force (Euler force is the least interesting one anyways).

§6.2 Centrifugal force

We have

$$\mathbf{F}_{\text{cen}} = -m\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{x})$$

so

$$|\mathbf{F}_{\text{cen}}| = m\omega^2 r\cos\theta$$

(DIAGRAM). Furthermore, the centrifugal force is conservative,

$$\mathbf{F}_{\text{cen}} = -\boldsymbol{\nabla}\left(-\frac{1}{2}m|\boldsymbol{\omega}\times\mathbf{x}|^2\right) = -\boldsymbol{\nabla}\left(-\frac{1}{2}m\omega^2 r^2\cos^2\theta\right)$$

which allows us to see that potential eenergy decreases moving away from the axis of rotation.

Example

Take a mass on a string hanging from a fixed point above the Earth at latitude θ. We want to calculate φ, the angle between the string and the vertical (DIAGRAM). Let us find the forces acting on the particle. We have

$$\mathbf{g} = -mg\hat{\mathbf{r}}$$

since the string is short compared to R_{earth}, we simplify by taking $\hat{\mathbf{r}}$ fixed. Also,

$$\begin{aligned}\mathbf{F}_{\text{cen}} &= -m\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{x}) \\ &= m\omega^2 r\cos\theta(\cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\boldsymbol{\theta}})\end{aligned}$$

(DIAGRAM). Then, to hold the string together, there must be a force exerted by the molecules in the string balancing the other forces – the tension $\mathbf{T}$. We can write (DIAGRAM)

$$\mathbf{T} = T\cos\varphi\hat{\mathbf{r}} + T\sin\varphi\hat{\boldsymbol{\theta}}$$

Now we have two equations for each of $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, and two unknowns T and φ. So the system should be soluble. We obtain

$$\tan\varphi = \frac{\omega^2 R\cos\theta\sin\theta}{g - \omega^2 R\cos^2\theta}$$

§6.3 Coriolis force

We have

$$\mathbf{F}_{\text{cor}} = -2m\boldsymbol{\omega}\times\mathbf{v}$$

where $\mathbf{v}$ is the velocity in the moving frame. Notably, this is the same as the Lorentz force for magnetic fields, but with $\mathbf{B}$ replaced with $-2\boldsymbol{\omega}$. We conclude that, in essence, the particle will move in a circle or spiral.

Example (Formation of hurricanes)

If a low pressure region forms in the atmosphere, then particles will move towards it – the Coriolis force bends them to follow.

DIAGRAM: particles going to a point, $-\boldsymbol{\omega}$ into the page at that point, Coriolis force going clockwise at each point by right-hand rule.

This, however, leads to a counter-clockwise swirling – hurricanes in the northern hemisphere rotate counter-clockwise, while those in the southern hemisphere rotate clockwise.

Also, nearer the equator, particles moving north or south are almost parallel to $\boldsymbol{\omega}$, so the Coriolis force is not so storng, explaining why hurricanes cannot form too near to the equator (although, for particles travelling east-west, it is very strong).

Example

Let us drop a ball from a tower of height h on the Equator (DIAGRAM). Where does it fall?

As the ball falls, its angular velocity must increase to preserve angular momentum, so the ball rotates faster than the Earth and hence falls in front of the tower. Let us perform these computations exactly using the Coriolis force (neglecting centrifugal since it acts in the same direction as gravity). Then

$$\begin{aligned} \ddot{\mathbf{x}} &= \mathbf{g} - 2\boldsymbol{\omega} \times \dot{\mathbf{x}} \qquad (*) \\ \implies \dot{\mathbf{x}} &= \mathbf{g}t - 2\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_0) \end{aligned}$$

Plugging this back into $(*)$, we have

$$\ddot{\mathbf{x}} = \mathbf{g} - 2\boldsymbol{\omega} \times \mathbf{g}t + 4\boldsymbol{\omega} \times (\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_0))$$

Now the final term is analogous to centrifugal force – it acts in the vertical direction and is small compared to gravity – so we neglect it. Then

$$\begin{aligned} \ddot{\mathbf{x}} &= \mathbf{g} - 2\boldsymbol{\omega} \times \mathbf{g}t \\ \implies \mathbf{x} &= \mathbf{x}_0 + \frac{1}{2}\mathbf{g}t^2 - \frac{1}{3}\boldsymbol{\omega} \times \mathbf{g}t^3 \end{aligned}$$

Now, to actually solve this, let us introduce a basis, with $\mathbf{e}_1$ pointing north, $\mathbf{e}_2$ pointing west, and $\mathbf{e}_3$ pointing up, so

$$\begin{aligned} \boldsymbol{\omega} &= \omega \mathbf{e}_1 \\ \mathbf{g} &= -g\mathbf{e}_3 \\ \mathbf{x}_0 &= (R+h)\mathbf{e}_3 \end{aligned}$$

so

$$\mathbf{x} = \begin{pmatrix} 0 \\ -\frac{1}{3}\omega g t^3 \\ R + h - \frac{1}{2}gt^2 \end{pmatrix}$$

The particle hits the ground at $\frac{1}{2}gt^2 = h$, and we can sub this in to find x_2, which will be negative – i.e. we moved in front of the tower as expected.

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Example 6.2 (Foucault's pendulum)

As the Earth rotates under Foucault's pendulum, the pendulum itself appears to rotate from the point of view of someone on Earth.

DIAGRAM. At a general latitude θ, we fix a basis $\mathbf{e}_1$ north along a tangent to the surface, $\mathbf{e}_2$ westward and $\mathbf{e}_3$ upward. Specify vectors

$$\begin{aligned}\mathbf{x} &= (x, y, z)\\ \mathbf{g} &= (0, 0, -g)\\ \boldsymbol{\omega} &= (\omega\cos\theta, 0, \omega\sin\theta)\end{aligned}$$

where $\mathbf{x}$ is the endpoint of the pendulum and $\boldsymbol{\omega}$ is the angular velocity of the Earth. Let l be the length of the pendulum, so

$$\mathbf{T} = \frac{T}{l}(-x, -y, l - z)$$

is the tension in the string (DIAGRAM). In particular, the inextensibility of the string gives that $x^2 + y^2 + (l - z)^2 = l^2$. It is now time to write down the equation of motion:

$$m\ddot{\mathbf{x}} = \mathbf{T} + m\mathbf{g} - 2m\boldsymbol{\omega} \times \dot{\mathbf{x}}$$

We have four equations and four unknowns, so the system should be soluble. The strategy is to solve the constraint to find z as a function of x and y, plug this into the ODE, eliminate T and solve the coupled ODEs. We will not do this here. The upshot is that the pendulum follows a slowly rotating ellipse in the x-y plane:

$$x + iy = e^{-i\omega t\sin\theta}\left(a\cos\left(\sqrt{\frac{g}{l}}t\right) + b\sin\left(\sqrt{\frac{g}{l}}t\right)\right)$$

Note that the period of rotation of the ellipse depends only on the angle θ.

§7 Special relativity

§7.1 Basic postulates of relativity

Maxwell's equations predicted the existence of electromangetic waves, which travel in vacuum at the speed of light

$$c = 299,792,458\text{ms}^{-1}$$

Any theory with a fixed velocity cannot have Galilean invariant, since in Galilean theory only relative velocities exist. Theoretically this could be OK – the same is true for sound – but we claim that, no matter one's own speed, light is still measured to travel at c.

Hence, by analogy with sound, light was assumed to be travelling in the so-called 'luminiferous ether'. But this was famously disproven by Michelson and Morley. Hence, Einstein built a new system based on the following postulates:

1. *(Principle of relativity)* The laws of physics are the same in all inertial frames.
2. The speed of light in a vacuum is the same in all inertial frames.

It is clear that Galilean invariance is incompatible with these postulates.

§7.2 Lorentz transformations

Let us for now consider only one spatial dimension x, and two frames with coordinates (x,t) and (x',t').

Recall that, according to Galileo,

$$\begin{aligned} x' &= x - vt \\ t' &= t \end{aligned}$$

since t cannot change between frames. But let us allow now for a more general

$$\begin{aligned} x' &= f(x,t) \\ t' &= g(x,t) \end{aligned}$$

The law of inertia still holds, so a free particle moves at a constant velocity. Hence, we know that, for such a particle,

$$\begin{aligned} x &= A + Bt \\ x' &= A' + B't' \end{aligned}$$

We will also choose that our frames have a constant origin, so $x = t = 0 \iff x' = t' = 0$. We see that the transformation must map all lines through the origin in the (x,t) plane to lines through the origin in the (x',t') plane – in other words, the transformation is linear. Hence

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

for a, b, c, d which don't depend on x, t, but can depend on the relative velocity between the frames.

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The frame S' is moving at speeed v in the frame S, but is at rest with respect to itself – hence, the line $x = vt$ should map to $x' = 0$ under the transformation. Hence, $x' = \gamma_v(x - vt)$ for some γ_v depending on v. Now we have some more steps to do.

1. First, we would like to show $\gamma_v = \gamma_{-v}$. Consider $\tilde{S}$ and $\tilde{S}'$ where the x-axis is inverted in direction, i.e. $\tilde{x} = -x$, $\tilde{x}' = -x'$.

   Now, if S' moves at speed v with respect to S, then $\tilde{S}'$ moves at speed $-v$ with respect to $\tilde{S}$. Hence

   $$\begin{aligned} &\tilde{x}' = \gamma_{-v}(\tilde{x} + vt) \\ \implies &x' = \gamma_{-v}(x - vt) \end{aligned}$$

   so, compaing with the definition of γ_v, we obtain the desired result.

   Another argument for this fact is that there is no preferred direction (by relativity), so γ should only depend on the magnitude of v (this will help us generalise to 3D).

2. If we boost by v and then by $-v$, we want to get back to the original frame. In equations,

   $$x = x'' = \gamma_{-v}(x' + vt') = \gamma_{-v}\left(\gamma_v(x - vt) + vt'\right)$$

$$= \gamma^2(x - vt) + \gamma v t'$$
$$\implies \boxed{t' = \gamma t + \frac{1-\gamma^2}{\gamma v} x}$$

Now we have x' and t' in terms of x and t, with γ the only undetermined quantity.

3. Use postulate 2, that the speed of light is the same in all inertial frames. That is, the light ray $x = ct$ must map to $x' = ct'$. Hence

$$\begin{aligned} x' = ct' &= c\left(\gamma t + \frac{1-\gamma^2}{\gamma v} x\right) \\ \implies \gamma(c - v)t &= c\left(\gamma + \frac{1-\gamma^2}{\gamma} \cdot \frac{c}{v}\right) t \\ \implies &\boxed{\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}} \end{aligned}$$

Notably, this transformation only makes sense for $v < c$. We have now derived the full form of a Lorentz transformation:

$$\begin{aligned} x' &= \gamma(x - vt) \\ t' &= \gamma\left(t - \frac{v}{c^2}x\right) \end{aligned}$$

For velocities $v \ll c$ (the *non-relativistic limit*), we would expect this to reduce to a Galilean transformation. We have that $\gamma \approx 1$, and $\frac{v}{c} \to 0$, and simply substituting gives $x' = x - vt$, $t' = t$ as expected.

Let us now explore the surprising consequences of this formulation.

§7.3 Addition of velocities

Suppose a particle is moving with speed u' in a frame S', itself moving at a speed v in frame S. What speed u does the particle move at in S? We will need to apply the inverse Lorentz transformation, which is unsurprisingly

$$\begin{aligned} x' &= \gamma(x + vt) \\ t' &= \gamma\left(t + \frac{v}{c^2}x\right) \end{aligned}$$

(we just changed the signs of v). Hence

$$\begin{aligned} u = \frac{x}{t} &= \frac{x' + vt'}{t' + \frac{v}{c^2}x'} = \frac{\frac{x'}{t'} + v}{1 + \frac{v}{c^2} \cdot \frac{x'}{t'}} \\ &= \frac{u' + v}{1 + \frac{vu'}{c^2}} \end{aligned}$$

We can easily see that, if v, $u \ll c$, this reduces to a simple addition, and u is also much less than the speed of light. Furthermore, this addition formula can never produce a velocity faster than the speed of light – for example, if $u' = v = c$, then

$$u = \frac{c + c}{1 + \frac{c^2}{c^2}} = c$$

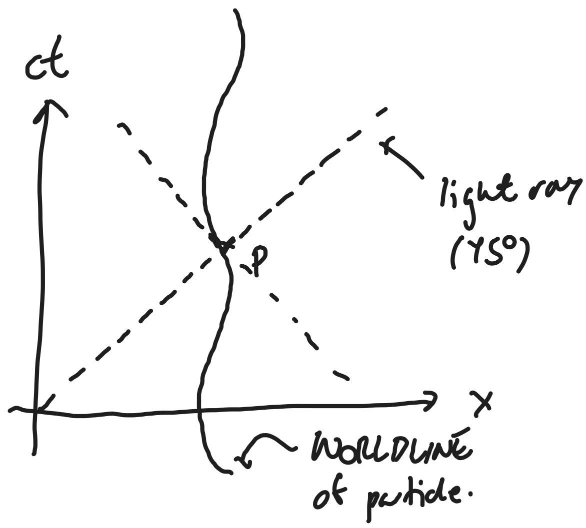

Figure 1: An example of a spacetime diagram.

§7.4 Spacetime diagrams and simultaneity

Note that light rays always travel at 45° on such a diagram by the choice of axes. Importantly, we can put the axes for a frame S' on the spacetime diagram for S. Since the t' and x' axes are at $x' = 0$ and $t' = 0$ respectively, we have

$$ct = \frac{c}{v}x$$
$$ct = \frac{v}{c}x$$

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giving: DIAGRAM. Drawing lines of constant t and t', we can see that events which are simultaneous in one frame may not be simultaneous in another – this is called the *relativity of simultaneity.* This is in fact an unavoidable consequence of the speed of light being constant, as exemplified in a thought experiment of Einstein.

DIAGRAM.

In the reference frame of the train, the light clearly reaches the back and front at the same time. On the other hand, in the reference frame of the platform, the back and front of the train have moved forwards, so the light has less far to travel to reach the back of the train, and further to reach the front – the light rays do not hit simultaneously.

In Galilean physics, on the other hand, light would be observed going at speed $c + v$ in one direction and speed $c - v$ in the other, reaching both ends at the same time.

One might worry that, if different frames see things at different times, one might be able to reverse cause and effect. However, this does not happen.

DIAGRAM (light cones)

The lines of simultaneity of S' can be at most at 45°, as $v \to c$. Hence, we can make events to the 'side' of P, such as R, simultaneous, but not events in either of P's light cones, such as Q or S. In other words, causality is the same in all frames.

§7.5 Time dilation

Suppose a clock at rest in frame S' ticks an intervals T' – so the ticks occur at $(0, 0)$, $(0, cT')$, $(0, 2cT')$, etc. In frame S, using inverse Lorentz transformations,

$$t = \gamma\left(t + \frac{v}{c^2}x'\right)$$

With $x' = 0$, ticks occur at $t = 0$, $t = \gamma T'$, etc – in other words, at intervals $T = \gamma T'$. Since $\gamma > 1$, $T > T'$, so S measures more time between the ticks of the clock in S' – a moving clock runs more slowly.

§7.6 The twin paradox

Twin A stays on Earth and twin B goes to Neptune and back at almost light speed. Both see the other twin moving while they remain at rest, so which one ends up younger?

The key asymmetry is that twin B has to turn around.

DIAGRAM.

From the perspective of A, time $2T$ passes, while, from the perspective of B, time $2T' = 2T/\gamma$ has passed.

DIAGRAM.

As we can see, while they are moving, they both see the other clock moving more slowly – so $2\hat{T} = 2T'/\gamma$ passes – but when B turns around, the simultaneity line changes, so A's clock skips some time.

§7.7 Length contraction

A rod has length L' at rest in S':

DIAGRAM.

Now, consider the moving frame S, in which the length of the rod is $P_1P_3 > P_1P_2$. How can we find the exact value of L in terms of L'?

In both frames, we have $P_1 = (0, 0)$, and P_2 is $(ct', x') = (0, L')$, so, taking the inverse Lorentz transformation, $P_2 = (\gamma\frac{v}{c}L', \gamma L')$ in S. Then, for P_3 (why??),

$$\begin{aligned}
x|_{P_3} &= x|_{P_2} - vt|_{P_2} \\
&= \gamma L' - v\frac{\gamma v}{c^2}L' \\
&= \gamma L'\left(1 - \frac{v^2}{c^2}\right) \\
&= \frac{L'}{\gamma}
\end{aligned}$$

so moving objects appear shorter. This is called *Lorentz contraction.*

Example

Consider a man carrying a ladder of length $2L$ running towards a barn of length L.

DIAGRAM (non-detailed)

From the point of view of a stationary observer, the ladder will appear to fit in the barn if A moves fast enough, since its length is $\frac{2L}{\gamma}$. On the other hand, in the frame of the perosn carrying the ladder, it is the barn that gets contracted, and so the ladder will definitely not fit in.

DIAGRAM (spacetime diagram)

§7.8 The invariant interval

Definition 7.1 (The invariant interval)**.** Consider two events P_1 and P_2 with coordinates (ct_1, x_1) and (ct_2, x_2) in frame S. Let

$$\Delta t = t_2 - t_1$$
$$\Delta x = x_2 - x_1$$

Then, the *invariant interval* Δs is given by

$$(\Delta s)^2 = c^2(\Delta t)^2 - (\Delta x)^2$$

Let us check it is invariant between frames. Observe

$$\begin{aligned} c^2(\Delta t')^2 - (\Delta x')^2 &= \gamma^2 c^2 \left(\Delta t - \frac{v}{c^2}\Delta x\right)^2 - \gamma^2(\Delta x - v\Delta t)^2 \\ &= (\Delta t)^2(\gamma^2 c^2 - \gamma^2 v^2) + (\Delta x)^2\left(\gamma^2 \frac{c^2 v^2}{c^4} - \gamma^2\right) \\ &= c^2(\Delta t)^2 - (\Delta x)^2 \end{aligned}$$

This is nice – observers in all frames can agree on the value of Δs. Furthermore, we can write

$$(\Delta s)^2 = \begin{pmatrix} c\Delta t & \Delta x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c\Delta t \\ \Delta x \end{pmatrix}$$

where the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is called the *Minkowski metric*. Using this approach, we can write Lorentz transformations in the form

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\frac{\gamma v}{c} \\ -\frac{\gamma v}{c} & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}$$

Then, invariance of Δs is equivalent to the fact that

$$\begin{pmatrix} \gamma & -\frac{\gamma v}{c} \\ -\frac{\gamma v}{c} & \gamma \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & -\frac{\gamma v}{c} \\ -\frac{\gamma v}{c} & \gamma \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

i.e., Lorentz transformations preserve the Minkowski metric.

> **Remark.** This is analogous to the fact that rotations preserve the Euclidean metric, namely I – that is,
>
> $$R\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}R^{-1} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

We should think of the Minkowski metric as measuring distances in spacetime. It is notably not positive-definite, so we have a distinction – points with

- $(\Delta s)^2 > 0$ are called *timelike separated*;
- $(\Delta s)^2 < 0$ are called *spacelike separated*;
- $(\Delta s)^2 = 0$ are called *lightlike separated*.

§7.9 Rapidity

Definition 7.2 (Rapidity)**.** We define the *rapidity* φ to be such that

$$\gamma = \cosh\varphi$$

Now nice things happpen – for example, $\sinh\varphi = \sqrt{\cosh^2\varphi - 1} = \frac{v\gamma}{c}$. Most nicely, the Lorentz transformation becomes

$$\begin{pmatrix}\gamma & -\frac{\gamma v}{c}\\ -\frac{\gamma v}{c} & \gamma\end{pmatrix} = \begin{pmatrix}\cosh\varphi & -\sinh\varphi\\ -\sinh\varphi & \cosh\varphi\end{pmatrix} = \Lambda(\varphi)$$

Particularly nicely, it holds that

$$\Lambda(\varphi_1)\Lambda(\varphi_2) = \Lambda(\varphi_1 + \varphi_2)$$

so rapidity is to a Lorentz transformation what angle is to a rotation. In contrast, in terms of velocities, addition is weird, namely

$$\Lambda(v_1)\Lambda(v_2) = \Lambda\left(\frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}\right)$$

as in the earlier addition of velocity formula.

§7.10 Lorentz transformations in four dimensions

Definition 7.3 (4D Minkowski metric)**.** The *4D Minkowski metric* is

$$\eta = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 0 & 0 & -1 & 0\\ 0 & 0 & 0 & -1\end{pmatrix}$$

We write this with indices as $\eta_{\mu\nu}$, where μ, ν are 0, 1, 2, 3 – conventionally, we use Greek letter indices for spacetime.

An event in spacetime is now given by a 4-vector,

$$X = (ct, x, y, z)$$

which will be denoted by capital letters with no underlines. We index with exponents, X^μ.

Now, the invariant distance between the origin and the event X is given by the inner product

$$\begin{aligned} X \cdot X &= X^T \eta X (= X^\mu \eta_{\mu\nu} X^\nu) \\ &= c^2t^2 - x^2 - y^2 - z^2 \end{aligned}$$

As before, we have the distinction between *timelike*, *spacelike* and *lightlike* or *null* vectors.

Then, the 4D Lorentz transformations are 4 by 4 matrices Λ such that

$$X' = \Lambda X$$

The defining feature of Lorentz transformations is that they preserve the Minkowski metric, i.e.

$$\Lambda^\intercal \eta \Lambda = \eta \qquad (\dagger)$$

Since Λ has 16 components, and both sides of $(\dagger)$ are symmetric, giving ten constraints, we expect to find six families of transformations. In particular, we have

- *Rotation and reflections*. These have

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}$$

  where $R^\intercal R = I_3$.

- *lorentz boosts* along the three possible axes. For example,

$$\Lambda_x = \begin{pmatrix} \gamma & -\frac{\gamma v}{c} & 0 & 0 \\ -\frac{\gamma v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  and similarly for other axes.

These matrices Λ form the Lorentz group, $O(1,3)$. Because the inner product is preserved by these transformations, it automatically takes lightlike vectors to other lightlike vectors, so that the speed of light is the same in all frames.

The subgroup with $\det \Lambda = 1$ is called the *proper Lorentz group*, $SO(1,3)$. A further subgroup are those transformations that do not reverse time, i.e. with top left entry positive, called the *proper orthochronous Lorentz group*, $SO^+(1,3)$. For example,

$$\Lambda = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

is in $SO(1,3)$ but not $SO^+(1,3)$.

§7.11 Proper time

We want to define a velocity to do physics with, but whose time should we differentiate with respect to? We should find a time τ that is invariant under Lorentz transformations, and define the 4-velocity as

$$U = \frac{dX}{d\tau}$$

Recall that the invariant interval $(\Delta s)^2$ is the same in all inertial frames. Hence, we define

$$\Delta\tau = \frac{\Delta s}{c}$$

Note that, since a worldline is timelike, $(\Delta s)^2 > 0$, so Δs is real. Now, we can parameterise the worldline as $x(\tau)$ and $t(\tau)$.

> **Remark.** This is analogous to the notion of arc length, but for worldlines.

Along a small section of the worldline, we have

$$\begin{aligned} d\tau &= \sqrt{dt^2 - \frac{d\mathbf{x}^2}{c^2}} \\ &= dt\sqrt{1 - \frac{1}{c^2}\left(\frac{d\mathbf{x}}{dt}\right)^2} \end{aligned}$$

Then, our 3-velocity $\mathbf{u} = \frac{d\mathbf{x}}{dt}$, so

$$d\tau = dt\sqrt{1 - \frac{\mathbf{u}^2}{c^2}} = \frac{1}{\gamma}dt$$

where γ is a function of the instantaneous 3-velocity $\mathbf{u}$. We have obtained

$$\frac{dt}{d\tau} = \gamma$$

For a clock following the worldline, $d\mathbf{x}' = \mathbf{0}$, so $d\tau = dt'$ – i.e., the proper time is the time measured by a (not necessarily inertial) observer following the worldline.

§7.12 4-velocity

Because τ is invariant and $X(\tau) = (ct(\tau), \mathbf{x}(\tau))$ transforms by Lorentz transformations, then

$$U = \frac{dX}{d\tau} = \frac{dt}{d\tau}\begin{pmatrix} c \\ \frac{d\mathbf{x}}{dt} \end{pmatrix} = \gamma\begin{pmatrix} c \\ \mathbf{u} \end{pmatrix}$$

which also transforms by Lorentz transformations (in fact, the defining property of a 4-vector is that it transforms by Lorentz transformations). In particular, we have

$$U \cdot U = \gamma^2 c^2 - \gamma^2 \mathbf{u}^2 = \gamma^2(c^2 - \mathbf{u}^2) = c^2$$

which is indeed invariant. Hence, we have a nice way of incorporating the familiar 3-velocity into a Lorentzian object.

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§7.13 4-momentum

Definition 7.4 (Relativistic energy and momentum)**.** The *relativistic energy* E and the *relativistic 3-momentum* $\mathbf{p}$ are defined by

$$E = m\gamma c^2$$
$$\mathbf{p} = m\gamma \mathbf{u}$$

Definition 7.5 (4-momentum)**.** The *4-momentum* of an object is given by

$$P = mU = \begin{pmatrix} m\gamma c \\ m\gamma \mathbf{u} \end{pmatrix}$$

where U is the 4-velocity, and m is a property of the particle called its *rest mass.*

Observe that the 4-vector P is

$$\begin{pmatrix} E/c \\ \mathbf{p} \end{pmatrix}$$

so the 4-momentum combines energy and momentum in the exact same way that the 4-position X combines time and space.

Proposition 7.6

In the absence of external forces, P is conserved, that is

$$\frac{dP}{d\tau} = 0$$

Remark. This is clearly a Lorentz-invariant generalisation of Newton's second law, as well as combining conservation of energy and momentum.

Proposition 7.7

The relativistic energy and momentum are related by

$$E^2 = \mathbf{p}^2 c^2 + m^2 c^4$$

Proof. Since $U \cdot U = c^2 \implies P \cdot P = m^2 c^2$,

$$\frac{E^2}{c^2} - \mathbf{p}^2 = m^2 c^2$$
$$\implies E^2 = \mathbf{p}^2 c^2 + m^2 c^4$$

■

In the non-relativistic limit, $\frac{|\mathbf{u}|}{c} \ll 1$, so $\mathbf{p} \approx m\mathbf{u}$, and we obtain the usual non-relativistic momentum. Furthermore,

$$E = m\gamma c^2 = \frac{mc^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$
\begin{aligned}
&= mc^2 \left(1 + \frac{1}{2}\frac{u^2}{c^2} + \dots\right) \\
&= mc^2 + \frac{1}{2}mu^2 + \dots
\end{aligned}
$$

and we can identify our familiar non-relativistic kinetic energy, as well as the *rest mass energy*, mc^2, a new consequence of relativity.

Let us now consider the high-velocity relativistic limit. Observe

$$\mathbf{p} = \gamma m \mathbf{u}$$

As $u \to c$, $\gamma \to \infty$, so the relativistic mass $\gamma m \to \infty$ as well. We end up needing more and more force to accelerate the particle by the same amount – and any finite force cannot accelerate the particle to the speed of light. This suggests that any particle with $m \neq 0$ should have $u < c$.

§7.14 Massless particles

In Galilean physics, massless particles don't really make sense – for example, the kinetic energy $\frac{p^2}{2m}$ would blow up. On the other hand, our relativistic expression suggests that

$$P \cdot P = m^2 c^2 = 0$$

but, since the inner product is not positive-definite, there are non-trivial solutions to this equation, namely those 4-momenta lying along a light ray. Indeed, for such a particle, 4-velocity does not exist, so 4-momentum really is the more useful notion.

A general light-like 4-momentum can be expressed as

$$P = \frac{E}{c}\begin{pmatrix} 1 \\ \hat{\mathbf{n}} \end{pmatrix}$$

where $\hat{\mathbf{n}}$ is a unit 3-vector. Interpreting $\hat{\mathbf{n}}$ as $\frac{\mathbf{u}}{c}$, where $\mathbf{u}$ is the 3-velocity, we obtain $\mathbf{u}^2 = c^2$, so massless particles move at the speed of light. We conclude that a particle is massless if and only if it moves at the speed of light.

There are some remarkable consequences of these results. For example, along a light ray $d\tau = 0$, because points on the trajectory are lightlike separated. This means that, for a massless particle, no time passes.

There are only two known massless particles, namely photons and gravitons. We will focus on photons, and we will have to borrow from quantum mechanics the formula that

$$E = \hbar\omega$$

where ω is the angular frequency of the light. Since frequency and wavelength are related, and length contraction causes a change in observed wavelength for different observers, the energy of photons may be perceived as different by different observers.

§7.15 Particle physics

Particle accelerators like CERN work by colliding particles at relativistic speeds. The proper framework for dealing with these collisions is quantum field theory, but we can still learn some things just from the fact that 4-momentum P is conserved. The two basic processes we will consider are particle decay and particle collisions.

§7.15.1 Particle decay

Heavy particles are often unstable, and decay into lighter particles – for example, the Higgs boson decays in about 10^{-22}s. Such particles must be detected by their decay products, for example

$$H \longrightarrow \gamma + \gamma$$

From the LHC, we know that $m_H c^2 \approx 125\text{GeV}$, about 10^5 times heavier than an electron. Conservation of 4-momentum tells us that

$$P_H = P_\gamma + P_{\gamma'}$$

Before we can proceed to split into components, we need to choose a frame to work in. There are two canonical options:

1. the *lab frame*, in which the particles are moving;
2. the *centre of mass* or *centre of momentum* frame, in which the total 3-momentum is zero.

For decays, 2 is just the rest frame of the starting particle, which makes life very easy.

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In the centre of mass frame, we have

$$P_H = \begin{pmatrix} m_h c \\ \mathbf{0} \end{pmatrix} \qquad P_\gamma = \frac{E_\gamma}{c}\begin{pmatrix} 1 \\ \hat{\mathbf{n}} \end{pmatrix} \qquad P_{\gamma'} = \frac{E_{\gamma'}}{c}\begin{pmatrix} 1 \\ \hat{\mathbf{n}}' \end{pmatrix}$$

Then, applying conservation, we obtain

$$\begin{aligned} m_H c &= \frac{E_\gamma}{c} + \frac{E_{\gamma'}}{c} \\ \mathbf{0} &= \frac{E_\gamma}{c}\hat{\mathbf{n}} + \frac{E_{\gamma'}}{c}\hat{\mathbf{n}}' \end{aligned}$$

from which it is clear that

$$\begin{aligned} E_\gamma &= E_{\gamma'} \\ \hat{\mathbf{n}} &= -\hat{\mathbf{n}}' \end{aligned}$$

Hence, two photons emerge with opposite 3-momentum and equal energy, namely half the rest mass of the Higgs boson – rest mass energy has been converted to kinetic energy.

Let us now try and do the same calculations in the lab frame. What angle θ does one of the emitted photons make with the path of the Higgs? (DIAGRAM)

We would like to use an *invariant* – for example, the inner product of two 4-vectors. Observe

$$\begin{aligned} P_H - P_\gamma &= P_{\gamma'} \\ \implies (P_H - P_\gamma)\cdot(P_H - P_\gamma) &= P_{\gamma'}\cdot P_{\gamma'} = 0 \\ \implies m_H^2 c^2 &= 2P_H \cdot P_\gamma \end{aligned}$$

Now, putting

$$P_H = \begin{pmatrix} E_h/c \\ \mathbf{p}_h \end{pmatrix} \qquad\qquad P_\gamma = \frac{E_\gamma}{c}\begin{pmatrix} 1 \\ \hat{\mathbf{n}} \end{pmatrix}$$

we obtain

$$m_H^2 c^2 = 2\left(\frac{E_H E_\gamma}{c^2} - \frac{E\gamma}{c}|\mathbf{p}_H|\cos\theta\right)$$

Given E_H and E_γ, we can solve for θ (since we also know $E_H^2 = p_H^2 c^2 + m_H^2 c^4$).

§7.15.2 Particle collisions

Suppose two particles of mass m collide and, in the centre of mass frame, scatter at an angle θ (DIAGRAM). Conservation of 4-momentum tells us that

$$P_1 + P_2 = P_3 + P_4 \qquad (\dagger)$$

so that, in the centre of mass frame, $\mathbf{p}_1 + \mathbf{p}_2 = 0 = \mathbf{p}_3 + \mathbf{p}_4$, and hence $|\mathbf{p}_1| = |\mathbf{p}_2|$, $|\mathbf{p}_3| = |\mathbf{p}_4|$. Furthermore, because the incoming particles have the same mass and momentum, they have the same energy, $E_1 = E_2$, and likewise for $E_3 = E_4$. Considering the time component of ($\dagger$), we hence obtain

$$\begin{gathered} E_1 = E_2 = E_3 = E_4 \\ |\mathbf{p}_1| = |\mathbf{p}_2| = |\mathbf{p}_3| = |\mathbf{p}_4| \end{gathered}$$

so that any angle θ is allowed, but the energy and momenta afterwards must equal the energy and momenta before. Since all momenta are in the same plane, we can furthermore specify, for example,

$$\begin{aligned} \mathbf{p}_1 &= |\mathbf{p}_1|(1,0,0) = -\mathbf{p}_2 \\ \mathbf{p}_3 &= |\mathbf{p}_1|(\cos\theta, \sin\theta, 0) = -\mathbf{p}_4 \end{aligned}$$

Lecturer then proceeded to introduce the standard model.

Let us now work in the lab frame, assuming one particle is at rest, and with scattering angles ϕ and α. (DIAGRAM)

Let's try and find ϕ. Once again, we must use invariants. We have

$$P_1 + P_2 = P_3 + P_4$$

We want to get rid of P_4, and we can do this by dotting it with itself, giving

$$\begin{aligned} m^2c^2 = P_4^2 &= (P_1 + P_2 - P_3)^2 \\ &= P_1^2 + P_2^2 + P_3^2 - 2P_1\cdot P_3 - 2P_2\cdot P_3 + 2P_1\cdot P_2 \\ &= 3m^2c^2 - 2P_1\cdot P_3 - 2P_2\cdot P_3 + 2P_1\cdot P_2 \end{aligned}$$

Since P_2 is at rest in the lab frame, it has the form

$$P_2 = \begin{pmatrix} mc \\ \mathbf{0} \end{pmatrix}$$

and we are now ready to substitute

$$P_1 = \begin{pmatrix} E_1/c \\ \mathbf{p}_1 \end{pmatrix} \qquad P_3 = \begin{pmatrix} E_3/c \\ \mathbf{p}_3 \end{pmatrix}$$

Hence

$$m^2c^2 = 3m^2c^2 - 2\left(\frac{E_1E_3}{c^2} - \mathbf{p}_1 \cdot \mathbf{p}_3\right) - 2E_3m + 2E_1m$$

and so, after rearrangement,

$$m^2c^2 + m(E_1 - E_3) - \frac{E_1E_3}{c^2} = -|\mathbf{p}_1||\mathbf{p}_3|\cos\phi$$

We have hence obtained $\cos\phi$ in terms of E_1, E_3 and $|\mathbf{p}_1|$, $|\mathbf{p}_3|$. Now we can simply apply

$$E_i^2 = \mathbf{p}_i^2c^2 + m^2c^4$$

to eliminate $|\mathbf{p}_i|$.

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§7.15.3 Particle creation

If two particles collide with enough energy, some of the energy can be used to create new particles. Let P_1 and P_2 have mass m, and the created particle have mass P_M. Starting from conservation of 4-momentum,

$$P_1 + P_2 = P_1' + P_2' + P_M$$

In the centre of momentum frame, $\mathbf{p}_1 = -\mathbf{p}_2$, and, since they have the same mass, $E_1 = E_2$. Hence, when adding them, the spatial components cancel, and we obtain

$$(P_1 + P_2) \cdot (P_1 + P_2) = \frac{4E_1^2}{c^2}$$

Meanwhile,

$$(P_1' + P_2' + P_M)^2 = \underbrace{P_1'^2 + P_2'^2 + P_M'^2}_{2m^2c^2 + M^2c^2} + 2(P_1' \cdot P_2' + P_1' \cdot P_M + P_2' \cdot P_M)$$

We will now show, as a lemma, that $P_1 \cdot P_2 \geq m_1m_2c^2$. Since $P_1 \cdot P_2$ is invariant, we can work in any frame. A convenient one happens to be in the rest frame of P_2, so

$$P_1 = \begin{pmatrix} E_1/c \\ \mathbf{p}_1 \end{pmatrix} \qquad P_2 = \begin{pmatrix} m_2c \\ \mathbf{0} \end{pmatrix}$$

$$\begin{aligned} \implies P_1 \cdot P_2 &= E_1m_2 \\ &= m_2\sqrt{p_1^2c^2 + m_1^2c^4} \\ &\geq m_1m_2c^2 \end{aligned}$$

Note we chose the plus sign of the energy – the other sign would lead to the wrong non-relativistic limit. In general we can always take energies to be positive when doing these calculations, and this is preserved by proper orthochronous Lorentz transformations.

We have

$$\begin{aligned}\frac{4E_1^2}{c^2} &= (P_1' + P_2' + P_M)^2 \geq 2m^2c^2 + M^2c^2 + 2(m^2c^2 + 2mMc^2) \\ &= 4\left(m + \frac{1}{2}M\right)^2 c^2 \\ \implies E_1 &\geq \left(m + \frac{1}{2}M\right) c^2 \\ \implies E_1 - mc^2 &\geq \frac{1}{2}Mc^2\end{aligned}$$

On the right hand side, we have approximately the kinetic energy of each incoming particle, and on the left hand side, we have half the rest mass energy of the created particle. So we are saying that P_1 and P_2 need to have enough kinetic energy to create the new particle.

If the inequality is saturated, then all kinetic energy is converted into rest mass energy, and the particles are stationary after the collision (in the centre of momentum frame).

§7.16 Accelerated motion

Definition 7.8 (4-acceleration)**.** The *4-acceleration* of an object is given by

$$A = \frac{dU}{d\tau} = \gamma \begin{pmatrix} \dot{\gamma} c \\ \dot{\gamma}\mathbf{u} + \gamma \mathbf{a} \end{pmatrix}$$

where $\mathbf{a} = \dot{\mathbf{u}}$ is the 3-acceleration.

> **Remark.** Note that, differentiating $U \cdot U = c^2$, we can see that $U \cdot A = 0$.

We want to solve motion for a constant acceleration – but constant in which frame? Take an inertial frame S' that, at some moment in time is instantaneously travelling at the same speed as the particle. Then, in this frame, the particle's velocity $\mathbf{u}' = \mathbf{0}$, so $\gamma = 1$ and $\dot{\gamma} = 0$.

Let us focus on two dimensions. We have

$$A' = \begin{pmatrix} 0 \\ a' \end{pmatrix}$$

We say the acceleration is constant when a' is constant – we will see next time that this is what happens when we have a constant force.

Given a' constant, what is a? Since A' is a 4-vector, we can take it to A with an inverse Lorentz transformation by speed u, which will of course give us a. In particular, we have

$$A = \begin{pmatrix} \gamma u a'/c \\ \gamma a' \end{pmatrix}$$

However, it is *not* the case that $a = \gamma a'$! Instead, we have

$$\gamma \dot{\gamma} c = \gamma u a'/c$$

$$\gamma(\dot{\gamma}u + \gamma a) = \gamma a'$$

and we have to solve the first equation for $\dot{\gamma}$, which we can sub into the second to obtain

$$a = \dot{u} = \left(1 - \frac{u^2}{c^2}\right)^{3/2} a'$$

which is a differential equation.

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We have

$$\int \frac{du}{a'\left(1 - \frac{u^2}{c^2}\right)^{3/2}} = \int dt$$

Setting $u = c\sin\theta$ to do the integral and recalling a' is constant, we obtain

$$u = \frac{a'ct}{\sqrt{c^2 + a'^2t^2}}$$

Hence, at early timees, $a't \ll c$, we have $u = a't$, while at late times, $a't \gg c$, so $u \to c$ – we cannot exceed c.

Since $u = \frac{dx}{dt}$, we can integrate again, to obtain

$$x = \frac{c}{a'}\sqrt{c^2 + a'^2t^2} + C$$

which is the equation of a hyperbola in spacetime – an accelerated particle follows such a hyperbola (DIAGRAM).

Observe that the accelerated particle never sees signals sent from x on the left of a certain light cone – this is called the *Rindler horizon*, separating the 'accessible' and 'inaccessible' regions of spacetime.

> **Remark.** A black hole's event horizon works similarly, since it is necessary to accelerate not to fall in.

Let us now check that the notion of instanteous acceleration as used earlier is rigorous.

Definition 7.9. The *force 4-vector* is given by

$$F = \frac{dP}{d\tau}$$

where P is the 4-momentum. Then Newton's second law is that

$$F = mA$$

We parameterise F as

$$F = \begin{pmatrix} F^0 \\ \gamma\mathbf{f} \end{pmatrix}$$

Let us now find the spatial component of this equation:

$$\frac{d\mathbf{p}}{dt} = \frac{1}{\gamma}\frac{d\mathbf{p}}{d\tau} = \frac{1}{\gamma}\gamma\mathbf{f} = \mathbf{f}$$

so $\mathbf{f}$ is actually just the normal force from the typical Newton's second law, even though $\mathbf{p} = \gamma m\mathbf{u}$ is not the typical momentum. Furthermore, $\mathbf{f}$ constant implies a' is constant in the instantaneous frame A, as we assumed earlier.

Now let us consider the time component: we have

$$F^0 = \frac{1}{c}\frac{dE}{d\tau} = \frac{\gamma}{c}\frac{dE}{dt}$$

so F^0 is proportional to the rate of change of energy. Indeed,

$$\begin{aligned} 0 &= \frac{d}{d\tau}P \cdot P = 2P \cdot \frac{dP}{d\tau} \\ &= 2\left(\frac{E}{c}\frac{dE}{d\tau} - \mathbf{p} \cdot \frac{d\mathbf{p}}{d\tau}\right) \end{aligned}$$

Recalling that $E = \gamma mc^2$ and $\mathbf{p} = \gamma mu^2$, we have

$$\begin{aligned} 0 &= 2\gamma^2 m\left(\frac{dE}{dt} - \mathbf{u} \cdot \frac{d\mathbf{p}}{dt}\right) \\ \implies \frac{dE}{dt} &= \mathbf{u} \cdot \frac{d\mathbf{p}}{dt} = \mathbf{u} \cdot \mathbf{f} \end{aligned}$$

which says that the change in energy is the rate of doing work – power is force times velocity.

Example (non-examinable)

We would like to show that the Lorentz force is Lorentz invariant. We claim it has the form

$$\frac{dP}{d\tau} = \frac{q}{c} G \cdot U \qquad (*)$$

where G is the electromagnetic 4-tensor, given by

$$G = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -cB_z & cB_y \\ E_y & cB_z & 0 & -cB_x \\ E_z & -cB_y & cB_x & 0 \end{pmatrix}$$

The time component of $(*)$ states that

$$\begin{aligned} \gamma \frac{dE}{dt} &= \frac{q}{c} \mathbf{E} \cdot \gamma \mathbf{u} \\ \implies \frac{dE}{dt} &= \frac{q}{c} \mathbf{E} \cdot \mathbf{u} \end{aligned}$$

i.e. the electric field does work (and the magnetic field does not). The spatial components of $(*)$ say that

$$\begin{aligned} \left(\gamma \frac{d\mathbf{p}}{dt} \right)^i &= \frac{q}{c} \left(G^{i0} U_0 - G^{ij} u_j \right) \\ &= \frac{q}{c} \left(E^i \gamma c + \varepsilon^{ijk} B^k c \gamma u_j \right) \\ \implies \frac{d\mathbf{p}}{dt} &= \frac{q}{c} \left(\mathbf{E} + \mathbf{u} \times \mathbf{B} \right) \end{aligned}$$

so the Lorentz force is indeed relativistically invariant, although only after accounting for the mixing of $\mathbf{E}$ and $\mathbf{B}$ under Lorentz transformations.

This concludes the course.